

# emgforge: an automated end-to-end pipeline for simulating surface EMG on MRI-based volume conductors

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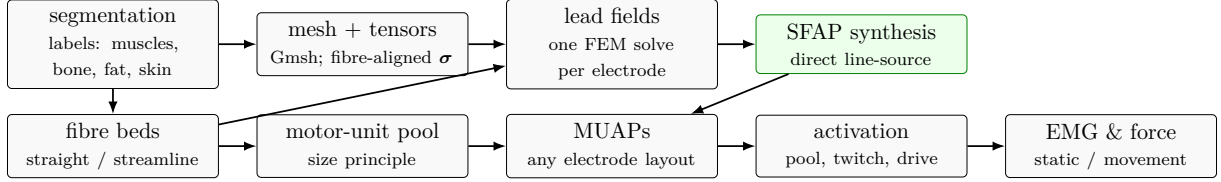
## Abstract

Simulated electromyograms are used to understand what an electrode records, to test decomposition and estimation algorithms on signals with known ground truth, and to train learning-based decoders. Most simulators fix the geometry to a cylinder or a slab, or stop at the lead field and leave the rest to the user. We present **emgforge**, an open pipeline that takes a labelled MRI segmentation of a limb to surface EMG in one command: a tetrahedral mesh with conductivity tensors aligned with each muscle’s fibres; one reciprocal finite-element solve per electrode, valid for every fibre of every muscle; fibre beds that are straight or follow the muscle’s shape; a motor-unit pool obeying the size principle; motor-unit action potentials on any electrode layout; and a motoneuron-pool and twitch layer that turns a drive or a movement into interference EMG and force. We describe the pipeline stage by stage, and at each stage we show, with an example, why the choice was made. The step that turns a lead field into a single-fibre action potential — direct line-source synthesis — is checked against a closed-form solution ( $r = 1.0000$ , zero lag), and the whole chain against fifty checks with numeric criteria from the physiological literature. We then use the pipeline for four studies: what an electrode sees as a function of depth, fat, spacing and montage; whether fibre geometry changes the signal; how much crosstalk a grid over one muscle receives from its neighbours; and how interference EMG scales with drive. The code, the validation suite and three datasets with ground truth are released.

## 1 Introduction

A forward model of the electromyogram (EMG) simulates the signal an electrode records when muscle fibres fire. Such models are used to understand what shapes a motor-unit action potential (MUAP), to test decomposition and estimation algorithms on signals whose ground truth is known, and, increasingly, to generate labelled training data for learning-based EMG decoding [1–3]. The classical models are analytical: a fibre in an infinite anisotropic medium [4–6], a planar layered conductor [7, 8], or a multilayer cylinder [9–11]. Finite-element models lift the geometric restrictions [12–14], and with reciprocity they run on MRI-derived anatomy at interactive speed: the myoelectric digital twin of Maksymenko et al. [1] is the reference example, and generative surrogates trained on such twins [2, 3] make the output cheap. Open code exists at the planar level [15] and as multi-domain finite-element frameworks [16]. What has been missing is an open pipeline that goes from a segmentation to EMG without hand steps, on any geometry a segmentation describes, with every stage exposed so it can be inspected, replaced and tested.

**emgforge** is that pipeline. Its input is a labelled segmentation (muscles, bones, fat and skin) and an electrode layout; its output is EMG on those electrodes, with the spike trains,



**Figure 1:** The *emgforge* chain. The lead field is the hinge: computed once per electrode and sampled along every fibre of every muscle; direct line-source synthesis turns each sampled lead field into an SFAP; fibres are summed per motor unit, and the activation layer turns spike trains into interference EMG and force.

motor-unit properties, lead fields and MUAPs that produced it. The stages are the ones every volume-conductor model has — mesh, conductivities, lead fields, fibres, motor units, action potentials, activation — but each is a checkable object with a small interface, so the same activation layer runs on an analytical cylinder, a parametric limb or an MRI forearm, and the same validation checks run on all three. We built it because three of our own efforts needed it: a decomposition benchmark whose synthetic ground truth must be inspectable [17], a model-informed decomposition that inverts a forward model and is only as good as that model [18], and neural-operator surrogates of the lead field that inherit whatever the finite-element stage gets wrong [19, 20].

The paper is written to be read in order. Section 2 gives the whole chain in one figure and one table of runtimes. Section 3 walks through it stage by stage; at each stage we state the decision we took and show, with an example, why. Section 4 is the validation: the synthesis step against a closed-form solution, the finite-element conductor against the analytical cylinder, and the MUAPs and interference signals against the quantitative literature. Section 5 uses the pipeline for four studies and states what each one teaches. Section 6 discusses what the pipeline is for, what it does not yet do, and what comes next.

The contributions are: (i) an automated, anatomy-general pipeline from segmentation to EMG with one entry point and a per-stage cost that scales with electrodes, not with fibres or contractions; (ii) a synthesis method, *direct line-source synthesis*, that reproduces a closed-form line-source solution exactly and stays usable on finite-element lead fields; (iii) a reusable validation suite of fifty literature-based checks; (iv) four studies with conclusions for people who design electrodes, choose fibre models or interpret interference EMG; (v) three released datasets with ground truth.

## 2 The pipeline at a glance

Figure 1 shows the chain. A segmentation is meshed and given conductivities (Sec. 3.1); one finite-element solve per electrode gives that electrode’s lead field over the whole limb (Sec. 3.2); fibres are laid through each muscle (Sec. 3.3) and grouped into motor units (Sec. 3.4); the lead field sampled along each fibre is turned into a single-fibre action potential (SFAP) and the fibres of a unit are summed into its MUAP on every electrode (Sec. 3.5); and a motoneuron pool driven by an excitation or a joint angle turns the MUAPs into interference EMG and force (Sec. 3.6). Everything downstream of the solve is NumPy; only the solve needs FEniCSx [21] and Gmsh [22].

**One command.** The whole chain runs cold from the committed segmentation with `python scripts/run_pipeline.py --n-mu 100` (defaults: the FCU, a  $5 \times 5$  grid at 10 mm; any muscle label, grid and spacing can be given), which writes the mesh, the lead fields, the pool, the MUAP tensor and a contraction with its spike trains, force and EMG, together with the wall time of each stage. Table 1 gives those times for the forearm used throughout this paper (subject WR, flexor

carpi ulnaris, a  $5 \times 5$  grid). The point of the table is the shape of the cost: the finite-element work is per electrode and is done once; fibres, units and contractions reuse it.

**Why reciprocity.** By the reciprocity theorem for a linear volume conductor [23], the potential at electrode  $e$  due to a unit current source at a point  $\mathbf{r}$  inside a muscle equals the potential at  $\mathbf{r}$  due to a unit current injected at  $e$ . So one solve with the source at the electrode gives the *lead field*  $\varphi_e(\mathbf{r})$  everywhere, and sampling it along a fibre path  $\mathbf{r}(z)$  gives  $\varphi_e(z)$ , the weight with which membrane current at  $z$  reaches the electrode. Every fibre, every motor unit, every muscle and every contraction reuse the same solve; only the number of electrodes costs finite-element work. This is the design decision that makes the rest cheap, and the one that lets the crosstalk study of Sec. 5.3 use the same 25 solves for five muscles. The interior-point reciprocity of our solver is checked directly (Sec. 4.2).

### 3 The pipeline, stage by stage

#### 3.1 Anatomy from a segmentation

The input is a label map: each muscle, the bones, and fat/skin. Each label is meshed into a tetrahedral domain with a thin skin shell (Gmsh), and each domain receives a conductivity: isotropic for fat, skin and bone, transversely isotropic for muscle with a ratio of 5 along the fibre [24, 25]. The muscle tensor is rotated onto that muscle’s own centreline tangent, estimated from per-slice centroids and a smoothing spline, so every muscle is anisotropic along its own axis. The conductivities follow Gabriel et al. [26] at  $\approx 100$  Hz (Table 2). The forearm used here (subject WR) meshes to  $\approx 47,000$  cells in 42 s from a  $320 \times 320 \times 60$  label volume (Table 1).

*Why fibre-aligned anisotropy matters.* In an infinite medium the anisotropy elongates the lead field along the fibres by exactly  $\sqrt{5}$ , and in a layered cylinder by 1.5 (checked in Sec. 4); a muscle whose tensor pointed along the limb rather than along its own fibres would spread every MUAP along the wrong direction. *Why a distinct skin layer.* The literature disagrees by  $2000\times$  on skin conductivity (Table 2), and a thin conductive skin layer changes the lateral decay of the potential [10]; the pipeline keeps skin as its own domain so the choice is a parameter, not a mesh rebuild.

For validation the pipeline also builds *parametric limbs* — layered discs (cancellous and cortical bone, muscle, fat, skin) extruded or lofted along  $z$ , circular or elliptical, optionally tapered — and carries a line-by-line Python port of the analytical four-layer cylinder of Farina et al. [11], whose lead field is obtained by inverting its transfer function. The validation cylinder uses the analytical values in both models (Table 2).

#### 3.2 Lead fields by reciprocity

For each electrode the pipeline solves the quasi-static conductivity equation

$$\nabla \cdot (\boldsymbol{\sigma}(\mathbf{r}) \nabla u) = -s(\mathbf{r}), \quad (\boldsymbol{\sigma} \nabla u) \cdot \mathbf{n} = 0 \text{ on } \partial\Omega, \quad (1)$$

with the source  $s$  a zero-mean Gaussian blob of width  $\sigma_s$  centred on the electrode (its mean is subtracted so the pure-Neumann problem is solvable), by GMRES/ILU in FEniCSx. Electrodes are placed on the skin by ray-casting a layout onto the mesh; the layout used in this paper is a regular  $5 \times 5$  grid with 10 mm inter-electrode distance over the flexor carpi ulnaris (FCU), rows along the arm and columns around it (measured 10.2 mm along, 10.0 mm across). On the forearm the 25 solves and the sampling of every field along the 637 fibres of the FCU bed take 18 s in all (Table 1); on the  $1.03 \times 10^6$ -cell validation cylinder a solve takes 12 s. Figure 3c,d shows one grid electrode’s lead field on a slice and along three fibres at increasing distance: it is a smooth peak whose width grows with distance, the shape a sum of a few point sources has, which is what the synthesis step later exploits.

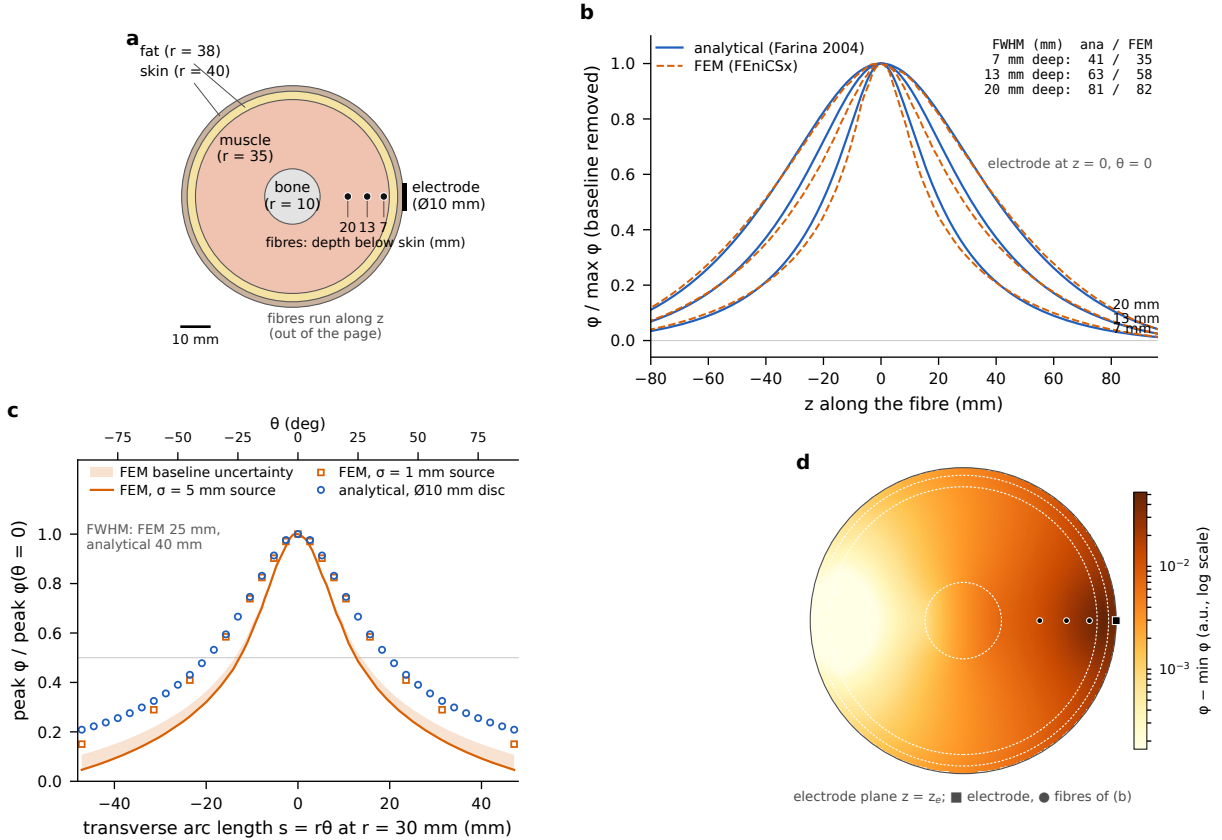
**Table 1:** One cold run of the pipeline from the committed forearm segmentation to interference EMG on the grid (subject WR, FCU,  $5 \times 5$  grid, 100 units, two worker processes on a desktop CPU shared with other jobs): what each stage does, its input and output sizes and its wall time. The finite-element work (mesh, tensors, 25 lead fields) is done in under a minute and once; the recipe’s monopole fits and the MUAP synthesis are per electrode–fibre pair and per unit and dominate; a contraction costs seconds.

| Stage                   | What it does                                                                                                                                                                                                                                    | Input $\rightarrow$ output                                                                         | Wall (s) |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------|
| Mesh                    | resample the labels, marching cubes on the tissue mask, smooth and decimate the surface, fTetWild tetrahedra, tissue tag per cell from the label under its centroid                                                                             | $320 \times 320 \times 60$ voxels $\rightarrow$ 46 749 tetrahedra, 11 022 nodes                    | 42.4     |
| Fibre directions        | PCA fibre direction and smoothed per-slice centreline of every muscle label $\rightarrow$ the fibre-aligned (5:1) conductivity tensor of each muscle                                                                                            | 21 labels $\rightarrow$ 17 muscles, 17 centrelines                                                 | 2.2      |
| Muscle geometry         | ray-cast boundary radius of each muscle cross-section per slice $\rightarrow$ the morphing-disk frame that carries a fibre through the muscle                                                                                                   | 17 centrelines $\rightarrow$ 17 cross-section models                                               | 6.0      |
| Fibre bed               | Poisson-disk sampling of the muscle cross-section at the set density; one curved morphing-disk path per fibre from tendon to tendon                                                                                                             | FlexorCarpiUlnaris, 206 mm <sup>2</sup> $\rightarrow$ 637 fibres $\times$ 200 points (204 mm long) | 1.3      |
| Motor-unit pool         | exponential Henneman size distribution; each unit’s territory grown around a random anchor fibre until it holds its fibres; index = size rank = recruitment order                                                                               | 637 fibres $\rightarrow$ 100 units of 5–395 fibres                                                 | 0.0      |
| Volume conductor        | conductivity tensor per cell from its tissue tag (muscle tensors rotated onto the centreline tangent), a skin shell inside the fat, FEniCSx spaces and search trees                                                                             | 46 749 cells $\rightarrow$ 46 749 $\sigma$ tensors, 11 022 dofs                                    | 5.3      |
| Electrode grid          | $M \times N$ grid ray-cast onto the skin surface over the muscle: rows IED apart along the arm, columns IED apart along the skin arc, centred on the limb-axis $\rightarrow$ muscle ray                                                         | 11 188 skin triangles $\rightarrow$ 25 electrodes ( $5 \times 5$ , $10.1 \times 10.0$ mm)          | 3.4      |
| Lead fields             | one reciprocity solve per electrode (zero-mean Gaussian source at the skin point, GMRES/ILU); $\varphi$ sampled along every fibre of the bed                                                                                                    | 25 solves $\rightarrow$ $\varphi$ bank<br>25 $\times$ 637 $\times$ 200                             | 17.8     |
| Lead-field conditioning | the recipe’s denoising: a free-position 3-monopole fit of each $\varphi(z)$ that removes mesh ripple before the second derivative — once per electrode–fibre pair                                                                               | 25 $\times$ 637 $\varphi(z)$ $\rightarrow$ 15 925 monopole fits                                    | 253.5    |
| MUAPs                   | per unit and electrode: line-source integral of the current-source density of a Rosenfalck action potential (both directions from the NMJ, one-sided tendon window) against $\varphi$ over the unit’s fibres; physical time, $t = 0$ at the NMJ | $\varphi + 100$ units $\rightarrow$ MUAP tensor $100 \times 25 \times 256$ (280 175 SFAPs)         | 658.6    |
| Activation              | motoneuron pool (recruitment thresholds, onion-skin rate coding, renewal ISIs) $\rightarrow$ spikes; twitch model $\rightarrow$ force; spikes convolved with the MUAPs $\rightarrow$ EMG on the grid, one trapezoid per drive level             | 6 trapezoids $\times$ 3.2 s $\rightarrow$ EMG 25 $\times$ 6554 per level, force, 32 858 spikes     | 1.8      |
| Total                   | sum of the stages above                                                                                                                                                                                                                         |                                                                                                    | 992      |

**Table 2:** Volume-conductor parameters (S/m). The validation cylinder uses the analytical values in both models; the production FEM/MRI table follows Gabriel et al. [26] at  $\approx 100$  Hz.

|                          | bone              | muscle $\sigma_{\perp}$ | muscle $\sigma_{\parallel}$ | fat    | skin                  | radii (mm)                         |
|--------------------------|-------------------|-------------------------|-----------------------------|--------|-----------------------|------------------------------------|
| analytical [11]          | 0.02              | 0.10                    | 0.50                        | 0.04   | 1.0                   | 10 / 35 / 38 / 40                  |
| FEM, validation cylinder | 0.02              | 0.10                    | 0.50                        | 0.04   | 1.0                   | 9, 10 / 35 / 38 / 40;<br>$L = 240$ |
| FEM / MRI, production    | 0.02 <sup>†</sup> | 0.2455                  | 1.228                       | 0.0379 | $4.55 \times 10^{-4}$ | from the<br>segmentation           |

<sup>†</sup>cortical; cancellous bone 0.075 S/m. The skin values differ by 2000 $\times$  between the two tables: the analytical school treats skin as a thin conductive layer (1 S/m), the Gabriel-based tables as dry skin ( $4.6 \times 10^{-4}$  S/m); it is the largest disagreement in the literature.



**Figure 2:** Lead fields on the validation cylinder. (a) Cross-section with the electrode and the fibre depths used. (b)  $\varphi(z)$  along fibres 7, 13 and 20 mm below the skin: analytical (solid) and FEM (dashed); the widths agree for deep fibres and differ for the shallowest, where the electrode models differ (a  $\varnothing 10$  mm disc against a Gaussian source that behaves as a  $\varnothing 5$  mm disc). (c) Transverse decay at the fibre depth ( $r = 30$  mm): the FEM field around the circumference (FWHM 25 mm; the band is the baseline uncertainty) against the analytical peak potential for fibres displaced around the circumference (FWHM 40 mm); with a  $\sigma_s = 1$  mm source the FEM profile matches the analytical one (extra points), so the difference is the 5 mm electrode blob, not the conductor. (d) The FEM lead field of one electrode on the cross-section (log scale); the dots are the fibres of (b).

*Why the source width is a decision.* A Gaussian source of  $\sigma_s = 5$  mm extends through 2 mm of skin and 3 mm of fat into the muscle, and its transverse footprint at the fibre depth is narrower than that of the analytical disc electrode (FWHM 25 against 40 mm, Fig. 2c); a 1 mm source reproduces the analytical profile. The width therefore sets the lateral footprint of every MUAP the pipeline produces, and it should be chosen to represent the physical electrode rather than for numerical convenience.

### 3.3 Fibre geometry

Two fibre models are available for any muscle in the segmentation. The *straight bed* places fibres on the muscle’s mid-length cross-section by Poisson-disk sampling at a chosen density (4 fibres/mm<sup>2</sup> for the FCU) and extends them as full-length paths parallel to the muscle centreline, with a junction at an atlas innervation-zone fraction (0.305 of the length for the FCU). This is the classical assumption and what cylinder models assume. The *harmonic-streamline bed* resolves the muscle’s internal architecture from its label mask alone: solve the masked Laplace problem  $\nabla^2\phi = 0$  inside the muscle with  $\phi = 0$  on the proximal cap,  $\phi = 1$  on the distal cap and an insulated surface, and trace the streamlines of  $\nabla\phi$  from seeds on the mid-belly cross-section. Streamlines follow the muscle’s shape, stay inside it and never cross, by construction; each is one full-length fibre with a single junction at the muscle’s shared innervation zone, and a quality guard rejects tracer failures (turn per step above 15°, or a longitudinal span below 30% of the muscle). An experimental variant cuts each streamline into short in-series fibres with their own junctions — the series-fibered that long muscles show [27]; it is gated behind a flag until validated.

*Why offer both.* On the FCU the two beds differ exactly where a straight model is weakest: the 827 streamlines (seeded at 0.5 mm) stay inside the mask for 99.96% of their length against 96.6% for the 637 straight fibres, and their lengths follow the muscle’s shape rather than one global value (Fig. 3b). Whether that changes the signal is a question, not an assumption, and Sec. 5.2 answers it. One convention to watch: the harmonic model measures its junction fraction along the muscle’s principal axis, which for the FCU points the other way from the straight bed’s fibre coordinate; the pipeline therefore innervates both beds on the same plane.

### 3.4 Motor-unit pool

Motor units are drawn from the bed following the size principle [28]:  $N$  sizes log-uniform between 5 and 400 fibres (many small units, few large), each unit anchored on a random bed fibre and its territory radius grown until it contains the target number of fibres, territories allowed to overlap; units are sorted by size so that index  $i$  is recruitment order. Because the bed is a fixed sample of the muscle (637 fibres for 11,207 fibre slots in the 100-unit pool), fibres are shared between units — 18 units per fibre on average — so a fibre’s SFAP is computed once and reused. Figure 4 shows the pool.

*Why a shared bed.* The alternative, one private set of fibres per unit, costs a lead-field sample and an SFAP per fibre slot (17× more) for no change in the sum: two fibres 0.5 mm apart have the same lead field to the precision of the mesh. The price is that the amplitude–size relation in Fig. 4d is a consistency check (with shared fibres at similar depths, amplitude must scale with fibre count) rather than evidence.

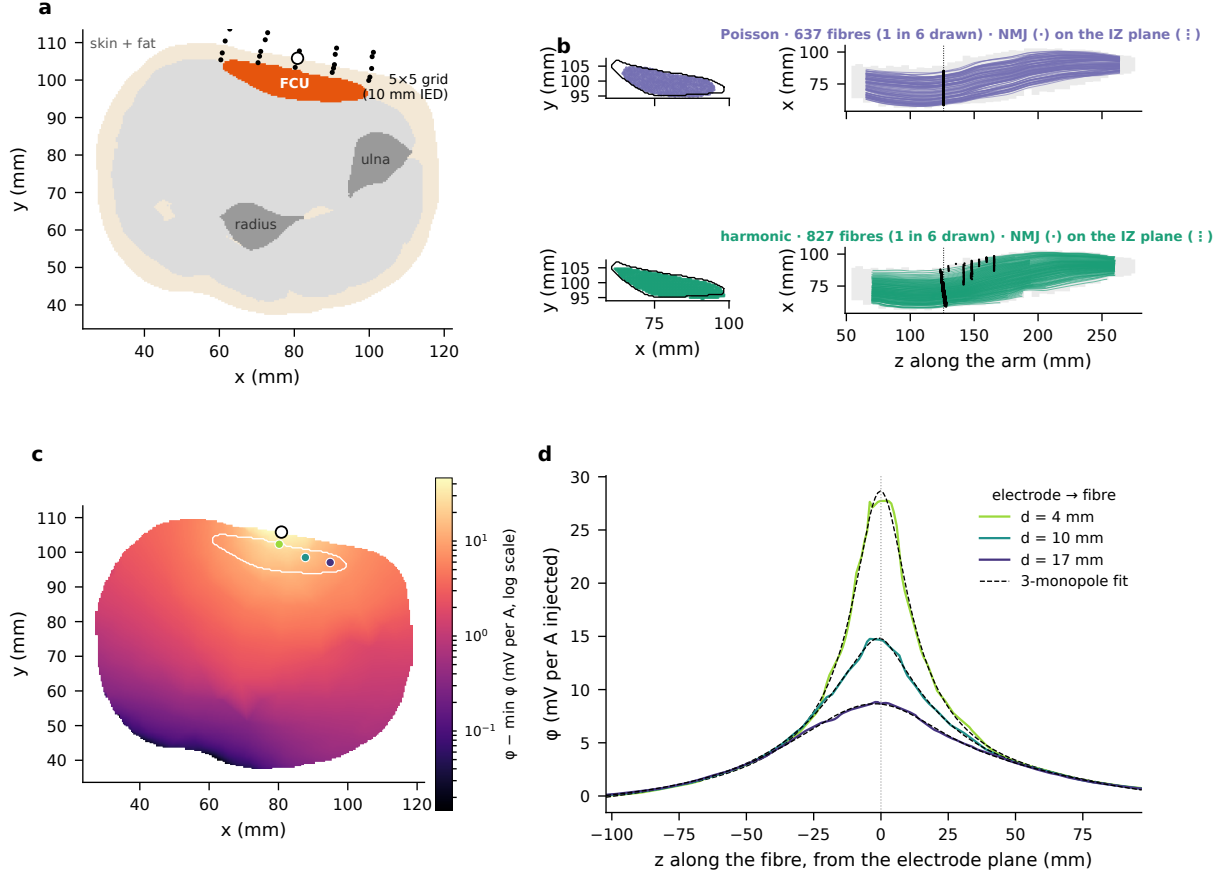
### 3.5 From lead field to action potential: direct line-source synthesis

We call the synthesis step *direct line-source synthesis*: the line-source integral is evaluated directly, in space, on the lead field sampled along the fibre. Its steps are few and each has a reason.

**The integral.** The action potential is the Rosenfalck form  $V(s) = A s^3 e^{-s}$  for  $s > 0$  ( $s$  in mm,  $A = 96$  mV) [4], launched from the neuromuscular junction at  $z_0$  in both directions and cut at the tendons by a window  $w(z)$  over  $[z_0 - L_1, z_0 + L_2]$ :

$$V_m(z, t) = V(vt - |z - z_0|), \quad V_m^w(z, t) = V_m(z, t) w(z), \quad (2)$$

so that  $t = 0$  is the instant the junction fires and the wave front reaches the proximal tendon at  $L_1/v$  and the distal one at  $L_2/v$ . A fibre of radius  $a$  and intracellular conductivity  $\sigma_{in}$  injects,



**Figure 3:** The MRI forearm (subject WR). (a) A segmentation slice with the FCU highlighted and the  $5 \times 5$  electrode grid projected. (b) The FCU fibre bed at 4 fibres/mm<sup>2</sup>: 637 straight Poisson fibres and 827 harmonic streamlines (one in six drawn), both innervated on the same plane (dotted; the atlas fraction 0.305 of the straight fibres' length): 84% of the streamlines cross it and are innervated within 1 mm of it; the rest start on its far side ( $z = 140$ – $165$  mm) and are innervated at their nearer end, the second cluster of dots. (c) The reciprocal lead field of one grid electrode on a mesh slice,  $\varphi - \min \varphi$  on a log scale (as in Fig. 2d). (d)  $\varphi$  along three fibres 4, 10 and 17 mm from the electrode, in mV per ampere injected, with the three-monopole fit the direct recipe uses (dashed).

per unit length, the membrane current of core-conductor theory [4, 29],

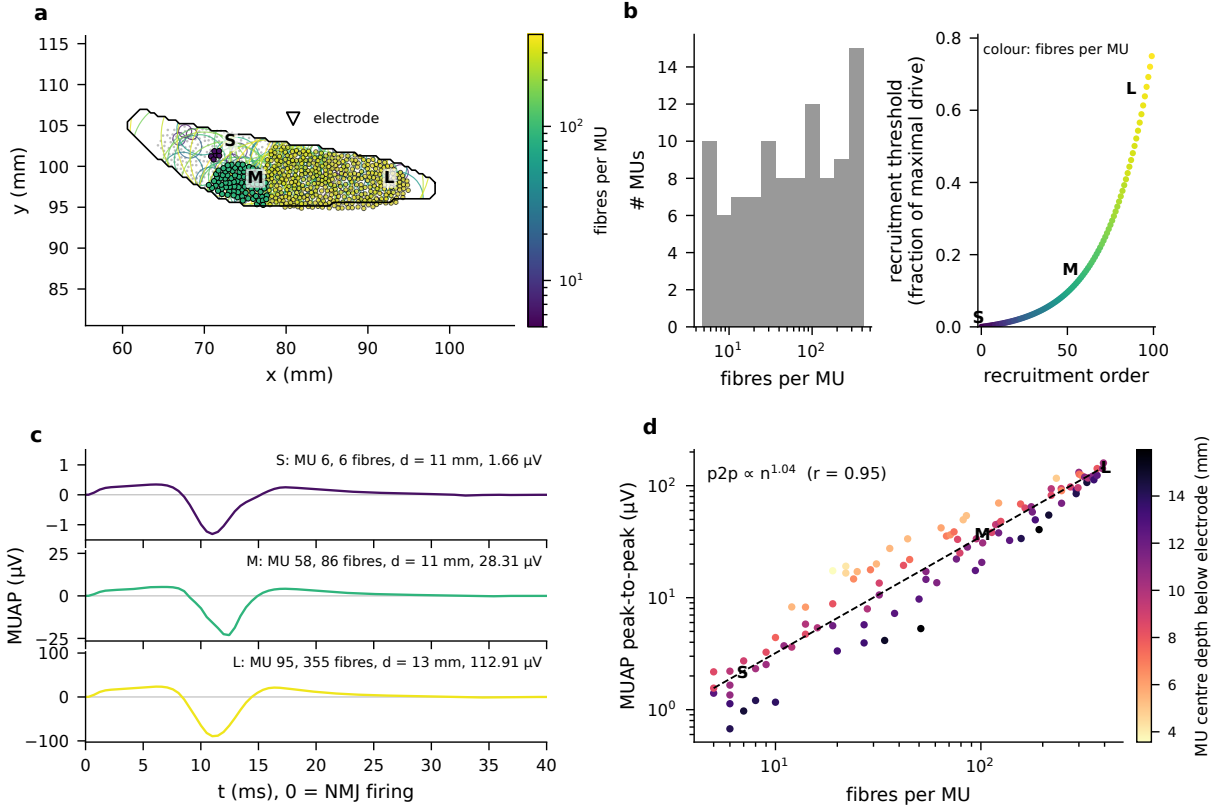
$$i_m(z, t) = \sigma_{in} \pi a^2 \frac{\partial^2 V_m^w}{\partial z^2}, \quad (3)$$

and the potential at the electrode is the lead-field-weighted sum of that current,

$$\text{SFAP}(t) = \int i_m(z, t) \varphi(z) dz = \sigma_{in} \pi a^2 \int V_m(z, t) w(z) \varphi''(z) dz, \quad (4)$$

where the second form follows by integrating by parts twice and shows that the SFAP is the inner product of the windowed action potential with the *second derivative* of the lead field. Taking  $\partial^2/\partial z^2$  of the whole windowed field, rather than of each travelling half analytically, produces the three source terms that the literature separates by hand [30, 31]: the two propagating tripoles, the *generation* term at the cusp  $|z - z_0|$ , and the *end-of-fibre* terms where the window cuts the wave. These are exactly the terms that make the source monopole-free,  $\int i_m dz = 0$  for all  $t$  [15]; we verify this to  $8 \times 10^{-17}$  (Sec. 4.1).  $\varphi$  from the finite-element stage is the potential in volts for 1 A injected at the electrode; the engine uses  $\sigma_{in} = 1$  S/m and  $a = 50 \mu\text{m}$ , so the constant  $\sigma_{in} \pi a^2$  is a scale factor rather than a calibrated quantity and absolute microvolt values carry that caveat.





**Figure 4:** The 100-unit FCU pool. (a) Territories on the cross-section, colour by size (5–395 fibres, median 66; territory radii 0.6–13.8 mm), with the centre electrode of the grid. (b) Size distribution, and recruitment threshold against recruitment order (colour: size). (c) MUAPs at the centre electrode of a small, a medium and a large unit (1.7, 28 and 113  $\mu\text{V}$ ; across the pool 0.7–160  $\mu\text{V}$ , median 24, median duration 18 ms). (d) MUAP amplitude against fibre count:  $p2p \propto n^{1.04}$  ( $r = 0.95$ ), the scatter being depth (colour); with shared fibres this is a consistency check (Sec. 3.4).

**The recipe.** Discretised, Eq. (4) is one matrix–vector product,  $\text{SFAP} = (\mathbf{C}\boldsymbol{\varphi})\Delta z$ , with  $\mathbf{C}$  the  $(n_t \times n_z)$  current-source-density matrix. The decisions around it, each with its reason, are (Fig. 5, Table 3):

1. **Monopole denoise of  $\boldsymbol{\varphi}$ .** A finite-element  $\boldsymbol{\varphi}(z)$  carries mesh-scale ripple, and the SFAP integrates  $\boldsymbol{\varphi}''$ , which amplifies that ripple quadratically in spatial frequency. Instead of low-passing — which rounds the real peak and halves the spectral content (Sec. 4.3) — we replace  $\boldsymbol{\varphi}$  by a least-squares fit of the form the field actually has,  $\hat{\boldsymbol{\varphi}}(z) = \sum_{i=1}^3 A_i / \sqrt{d_i^2 + (z - z_i)^2} + c$ , a sum of three free-position point sources. On a clean analytical  $\boldsymbol{\varphi}$  the fit is a near no-op ( $r > 0.99$ ); on a FEM  $\boldsymbol{\varphi}$  it changes the field by at most 1.8% of its peak but lifts the correlation of  $\boldsymbol{\varphi}''$  with the analytical one from 0.47 to 0.91 and takes the SFAP’s jaggedness from 0.011 to 0.002 (analytical: 0.0018; Fig. 5d,e).
2. **Edge taper.** A short cosine ramp (5 and 10 samples) on the two ends of the sampled  $\boldsymbol{\varphi}$ , a Gibbs guard for lead fields truncated by the sampling window.
3. **Upsampling  $\times 2$  (cubic spline).** The second derivative needs a finer grid than the raw  $\boldsymbol{\varphi}$  or the SFAP develops a Nyquist zigzag;  $\times 2$  removes it and  $\times 4$  adds nothing.
4. **Current-source density.**  $\mathbf{C}$  is built from the full bidirectional windowed field of Eq. (2) by two numerical  $z$ -derivatives, so the generation and end-of-fibre terms come out of the same operation as the tripoles.



**Table 3:** The direct line-source recipe (`SpatialConfig`). The cylinder tier uses  $f_s = 4096$  Hz; the MRI tier 2048 Hz with  $v = 3.3\text{--}4$  m/s.

| parameter                             | value                      | why                                                                                   |
|---------------------------------------|----------------------------|---------------------------------------------------------------------------------------|
| <code>denoise</code>                  | monopole, 3 poles          | fit the analytic form of $\varphi$ ; removes mesh ripple before $\varphi''$           |
| <code>edge_taper_left/right</code>    | 5 / 10 samples             | Gibbs guard at the sampling-window ends                                               |
| <code>upsample_factor</code>          | 2                          | second derivative off the Nyquist zigzag                                              |
| <code>csd_derivative</code>           | 2                          | physical current source (triphasic)                                                   |
| <code>fiber_window</code>             | one-sided, $\alpha = 0.25$ | flat at the junction, tapered at the tendons                                          |
| <code>center_time / t_start_ms</code> | false / $-10$              | physical time, junction at $t = 0$                                                    |
| $v, f_s, w$                           | 4 m/s, 4096 Hz, 256        | coupled grid $\Delta z = v/f_s$                                                       |
| $\sigma_{\text{in}}, a$               | 1, 0.05 mm                 | amplitude constant $\sigma_{\text{in}}\pi a^2$ ; no $1/v$ (the IAP is fixed in space) |

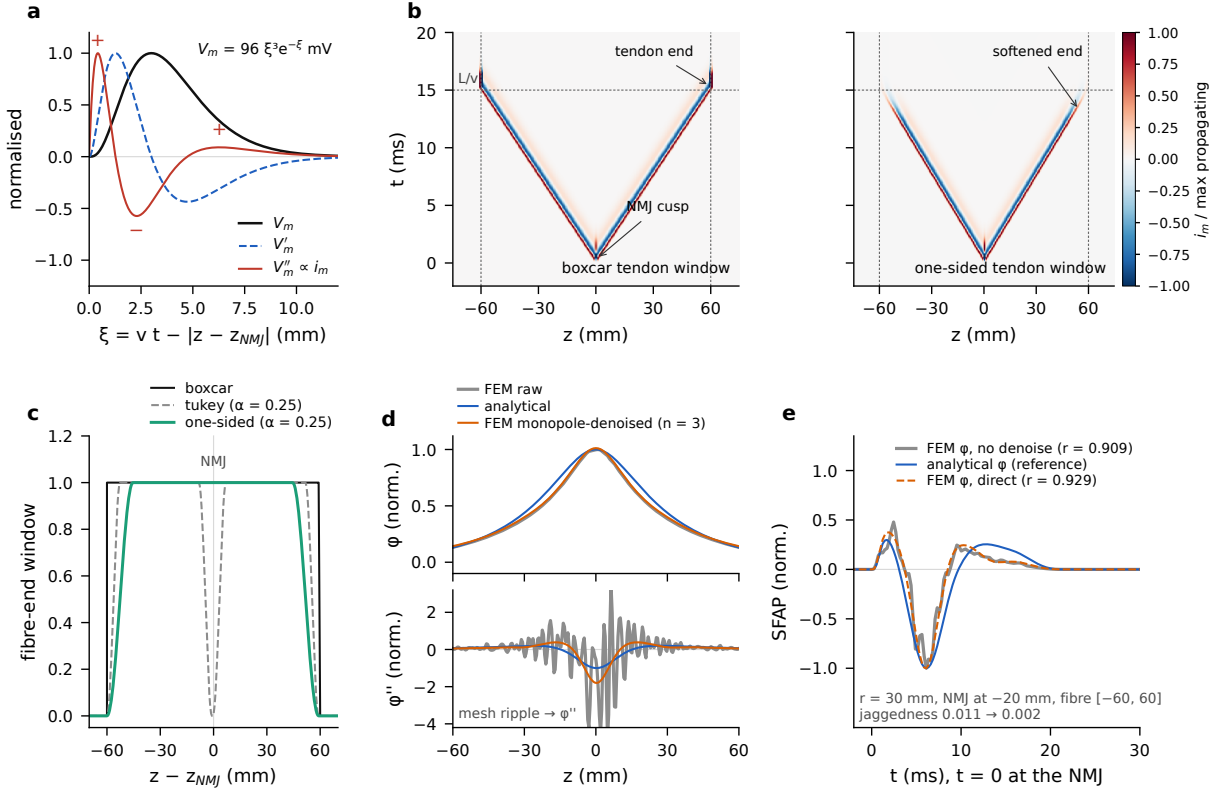
- 5. One-sided tendon window.**  $w(z)$  is flat through the junction and cosine-tapered over the last 25% of each semi-fibre only. A symmetric window would notch the junction, where the wave is born; a hard boxcar produces a sharp end-of-fibre spike (right for one fibre with a sharp tendon, and the choice we use against sharp-tendon oracles) which in real muscle is softened by tendon scatter (Fig. 8c). The taper costs  $r = 0.998$  against the sharp-tendon oracle.
- 6. Physical time.** No centring:  $t = 0$  is the junction, the detector lobe of an electrode  $\Delta z$  from the junction lands at  $\Delta z/v$ , the end-of-fibre onset at  $L/v$  (Fig. 8a); a 10 ms pre-roll is prepended so a junction under the electrode is not pinned to the window edge. Because time is physical, junction scatter, conduction-velocity scatter and tendon scatter across the fibres of a unit produce the temporal dispersion real MUAPs show rather than a window artefact: a 5 mm junction scatter across 50 fibres lengthens a MUAP from 20.0 to 22.2 ms and reduces its peak-to-peak to 0.86 of the coherent sum.

**Why not the spatial-frequency route.** The pipeline also carries a spatial-frequency-domain engine after Farina and Merletti [8], Farina et al. [11] —  $\varphi$  transformed along  $z$ , multiplied by the fibre-end function, the action-potential spectrum and  $jk_z$ , brought back to time by a Radon section — and a line-by-line port of the Farina generator. It was our first engine and the reference the direct method was originally tuned against. Given the same  $\varphi$ , our port does not reproduce the line-source integral: it is anti-phase and lagged by 0.9–2.6 ms while its amplitude agrees within 5% (Sec. 4.1), and in the ported generator the end-of-fibre onset leads  $L/v$  by 2.5 ms and the propagating main lobe has the opposite sign to the textbook convention. We attribute these to the orientation and sign conventions of the action potential in our port — the original model was verified against limiting cases and reproduced by finite elements to 3–5% [1] — and until that audit is done the direct method carries production and the other two are kept for comparison.

**From SFAPs to a MUAP.** A *fibre bed* object carries, per fibre, the sampling step (arc length for curved fibres), the two semi-lengths, the junction position and the conduction velocity; a MUAP is the sum of the SFAPs of the fibres in a motor unit, each computed from its own  $\varphi(z)$  with the shared configuration, on every electrode of the layout.

### 3.6 Activation: from motor units to signals

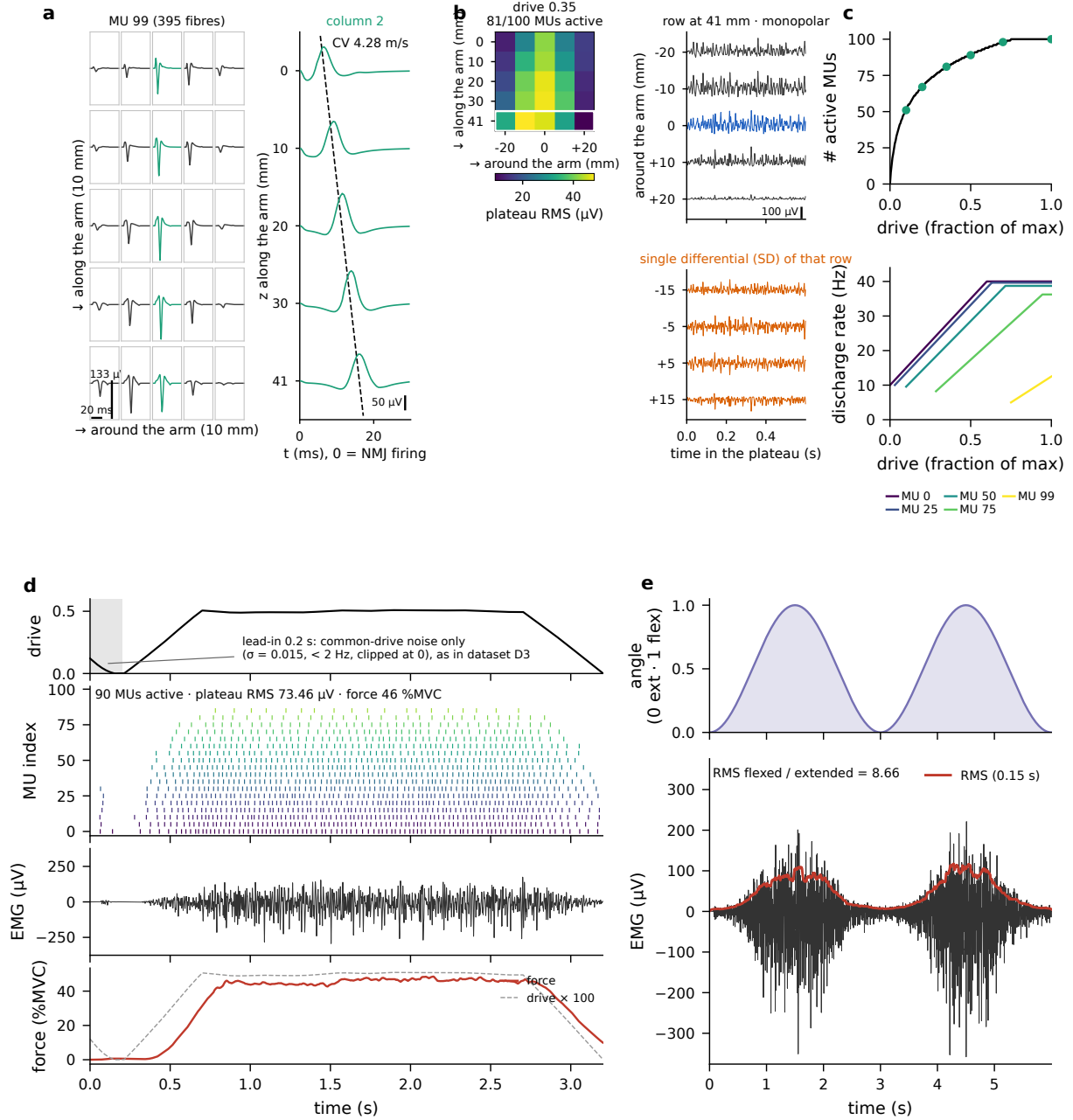
The activation layer is phenomenological, written from the equations of the Fuglevand pool as parametrised in NeuroMotion [3, 28]. Given a scalar excitation  $E(t) \in [0, 1]$ : unit  $i$  has a



**Figure 5:** Anatomy of direct line-source synthesis. (a) The Rosenfalck action potential and its first and second spatial derivatives; the second derivative is the triphasic current source. (b) The current-source-density matrix  $C(z, t)$  for a fibre  $[-60, 60]$  mm with the junction at 0: two tripoles leave the junction, the cusp at the junction and the window edges at the tendons appear as sources; boxcar (left) and one-sided (right) tendon windows. (c) The three fibre-end windows. (d) A finite-element lead field raw, monopole-denoised and analytical, with their second derivatives (normalised to the analytical peak; the colour scale in (b) is clipped at the propagating maximum — the tendon term is  $2.7\times$  larger). (e) The SFAPs each produces: the ripple in  $\phi''$  reaches the raw SFAP and the monopole fit removes it ( $r$  against the analytical- $\phi$  SFAP  $0.909 \rightarrow 0.929$ ); the remaining difference from the analytical waveform is the FEM field’s own shape at this shallow depth (Sec. 4.2), not ripple.

recruitment threshold spread exponentially with range 50 up to  $E = 0.75$  for the last unit; once recruited its rate is  $\min(PFR_i, MFR_i + (E - RTE_i) s_i)$  with the peak rate falling from 40 to 30 pps, the minimum from 10 to 5 pps and the slope from 50 to 30 pps per unit drive across the pool — the onion skin [32]; spikes are a renewal process with Gaussian inter-spike intervals of coefficient of variation  $1/6$ , re-drawn at each discharge, and a common-drive fluctuation (low-passed noise at 2 Hz) can be added to  $E$ . Each discharge adds a twitch  $P_i(t/T_i) e^{1-t/T_i}$  with amplitudes spanning  $100\times$  and larger units contracting faster, scaled by the rate-dependent fusion gain of Fuglevand et al. [28]; the sum is normalised to the mean force at full excitation, so force is in %MVC. Each spike places the unit’s MUAP (single channel or on every electrode) and the trains sum; single- and double-differential montages are differences of electrodes. For *movement*, a normalised joint angle  $\theta(t)$  sets the drive ( $0.08 + 0.42 \theta$ ), scales each spike’s MUAP ( $1 + g_a(\theta - 0.5)$ ; shortening brings fibres to the electrode) and time-warps it ( $1 + g_w(\theta - 0.5)$ ), with defaults  $g_a = 0.7$ ,  $g_w = 0.18$ .

*Why phenomenological.* The pool is the best-tested part of the EMG literature and its parameters are published; a biophysical motoneuron model would add cost and free parameters without a validation target at the surface. The movement layer is a modulation, not a musculoskeletal model; models of the geometry change itself exist [33, 34] and are the next step. Figure 6 shows a unit on the grid, an interference plateau, the pool’s recruitment and rate



**Figure 6:** From motor units to signals (regular  $5 \times 5$  grid, 10 mm spacing). (a) The largest unit's MUAP on the grid — rows run along the arm, columns around it — and, right, the column with the largest amplitude as a waterfall offset by electrode position: the wave walks down the column at 4.28 m/s (dashed guide; 4.1–4.8 over the pool, set 4.0). (b) Interference HD-EMG at a 35%-drive plateau (81 of 100 units active): the RMS map of the grid (9–48  $\mu$ V), one row of monopolar traces and the single-differential traces of that row at a shared scale. (c) Recruitment and rate coding of the pool. (d) A trapezoid contraction at half drive: drive (the lead-in carries the common-drive fluctuation), spikes, EMG (plateau RMS 73  $\mu$ V), force (46% MVC); across drives 0.1–1.0 the plateau RMS is 14–133  $\mu$ V and the force 7–100% MVC. (e) A movement trial: angle and non-stationary EMG (RMS flexed/extended 8.7).

coding, a trapezoid contraction with its force, and a movement trial.

**Table 4:** Validation scoreboard (branch `paper/arxiv-v1`; tiers C and S run on the released forearm pool, dataset D2). “pass / total” counts every check; “known” are documented limitations that fail without failing the tier; the two genuine failures are discussed in Sec. 6. Appendix A lists all 50 checks.

| tier | scope                                                 | pass / total | known | fail                                            |
|------|-------------------------------------------------------|--------------|-------|-------------------------------------------------|
| A    | cylinder: first principles, analytical, FEM, pipeline | 16 / 20      | 3     | A1.1b: FEM $\varphi''$ at mid-depth, $r = 0.93$ |
| B    | MUAP features vs literature                           | 14 / 14      | 0     | —                                               |
| C    | interference EMG and pool                             | 6 / 8        | 1     | C2: no refractory floor                         |
| S    | chain-level sanity                                    | 8 / 8        | 0     | —                                               |

## 4 Validation

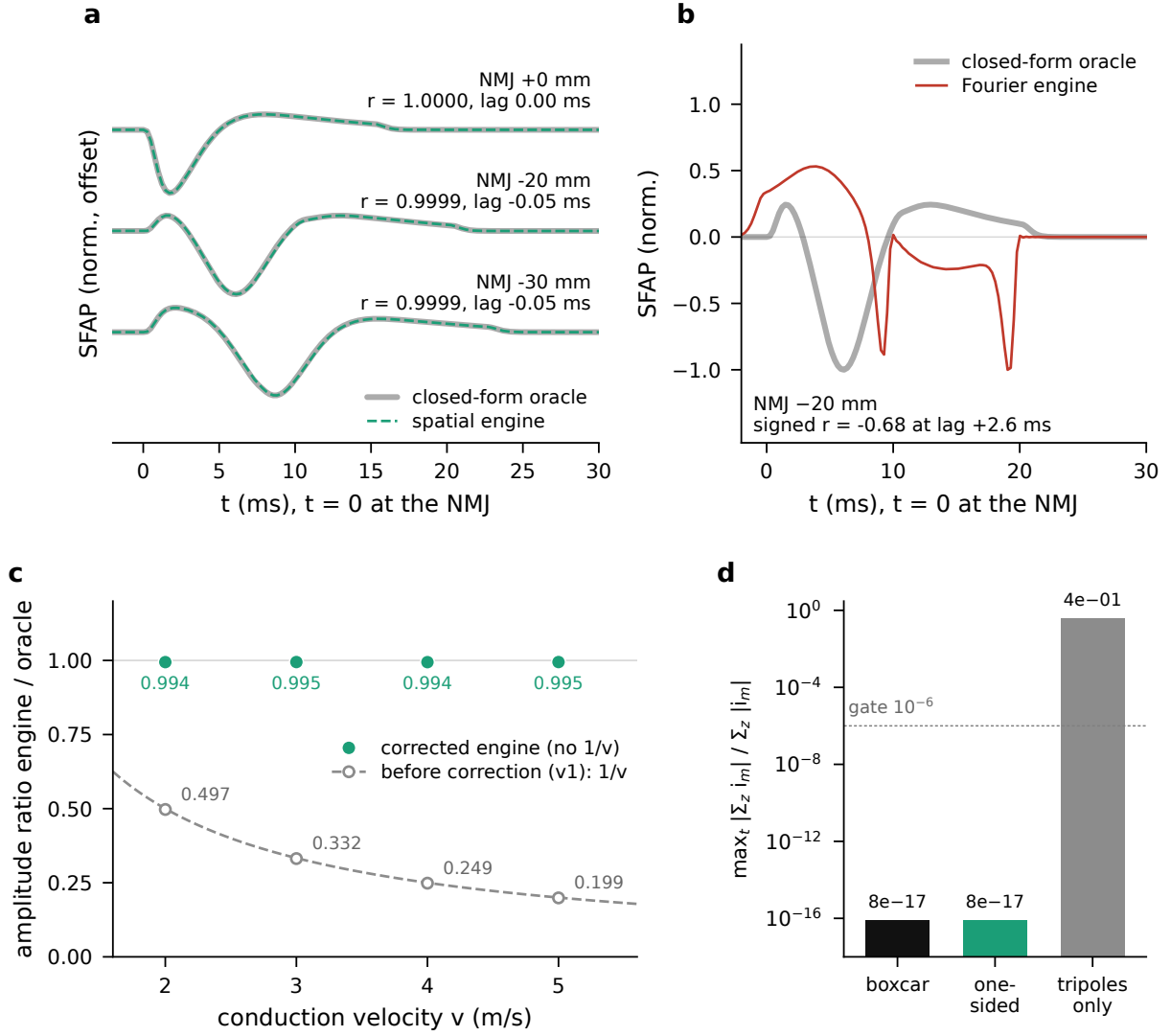
Every stage above can be tested in isolation, and the suite that ships with the code (`scripts/validation/`,  $\approx 4$  min) does so: fifty checks, each holding one measurable property against a literature principle with a numeric criterion, in four tiers — (A) the cylinder must reproduce: first principles, then the analytical model, then the finite-element model, then the pipeline on each; (B) the MUAP phenomenology the literature agrees on; (C) the statistics of interference EMG and the pool; (S) chain-level sanity checks. Table 4 is the scoreboard and Appendix A lists every check with its criterion and measured value. A “known” flag marks a documented limitation that keeps its criterion, so it turns green when fixed without failing the run. This section gives the results that matter for a reader of the pipeline; the open items are collected in Sec. 6.

### 4.1 The synthesis step against first principles

A fibre in an infinite anisotropic medium has a closed-form lead field,  $\varphi(z) \propto (\rho^2/\sigma_\rho + z^2/\sigma_z)^{-1/2}$  [25, 29], whose second derivative is analytic. Evaluating Eq. (4) in the dual form on a 0.02 mm grid gives an oracle with no numerical derivative and no volume-conductor model in the loop. Direct line-source synthesis, given the same  $\varphi$  on its 0.977 mm grid, matches it with  $r = 1.0000$ , 0.9999, 0.9999 and lags 0.00,  $-0.05$ ,  $-0.05$  ms for junctions 0,  $-20$ ,  $-30$  mm from the electrode (Fig. 7a). The source is monopole-free to  $7.7 \times 10^{-17}$ ; the SFAP is invariant to the sampling rate and step ( $r > 0.998$ , amplitude within 0.3%); moving the electrode by 20 mm delays the propagating lobe by 4.88 ms ( $\Delta z/v = 5.00$ , one sample); superposition is exact; and changing  $v$  from 4 to 3 m/s stretches the SFAP by 4/3 and scales its mean frequency by 3/4 [35]. The oracle also caught a defect: the first version of the engine divided by  $v$  once too often (a constant inherited from the spatial-frequency formulation, where the  $1/v$  belongs), so its ratio to the oracle was exactly  $1/v$ . With the division removed the ratio is 0.994 at every  $v$  from 2 to 5 m/s (Fig. 7c). The spatial-frequency engine on the same  $\varphi$  scores  $r = -0.79$ ,  $-0.68$ ,  $-0.64$  at lags of 1.4, 2.6, 0.9 ms (Fig. 7b).

### 4.2 The finite-element conductor against the analytical cylinder

Along fibres 7–20 mm below the skin the FEM  $\varphi(z)$  correlates with the analytical one at  $r = 0.995$ – $1.000$ , and at  $r = 0.997$ – $1.000$  for fibres 10–45° off the electrode meridian. Their widths differ for the shallowest fibres (FWHM 41 vs 35 mm at 7 mm), where the FEM Gaussian source behaves as a 5 mm-diameter disc; at mid-depth (13 mm)  $\varphi''$  agrees only to  $r = 0.93$  whatever electrode is assumed — the one tier-A failure. The pipeline run on FEM  $\varphi$  against the pipeline run on analytical  $\varphi$  gives  $r = 0.90$ – $0.97$  for shallow fibres and 0.998 for deep ones. Timing and symmetries hold: the end-of-fibre onset lands at 10.26 and 15.39 ms for  $L/v = 10$  and 15; conduction velocity from a four-electrode array is 3.98 m/s for  $v = 4$ ; rotating the electrode by 30° equals rotating the fibre ( $r = 1.000$ ); shifting it 20 mm along the axis equals shifting the

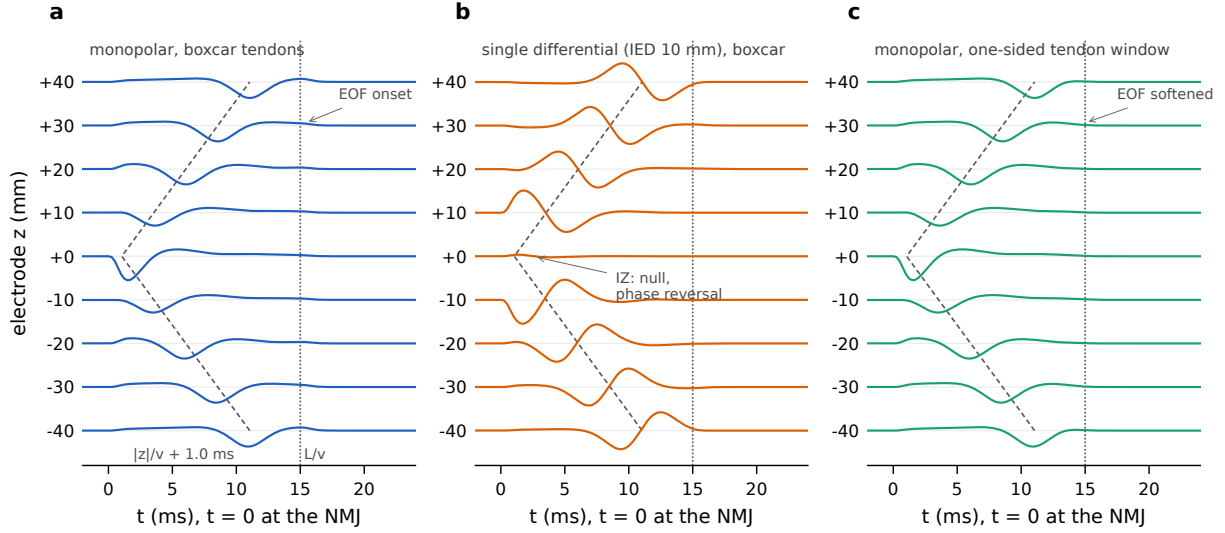


**Figure 7:** First principles. (a) The closed-form line-source SFAP and direct line-source synthesis, three junction positions. (b) The spatial-frequency engine against the same oracle. (c) The engine-to-oracle amplitude ratio: 0.994–0.995 at every  $v$  after the correction described in the text; the dashed  $1/v$  is what the uncorrected engine gave. (d) The net source current at every instant is zero to machine precision once the junction and end terms are included; the propagating tripoles alone leave 39%.

sampling window ( $r = 0.996$ – $0.999$ , the residual being the 240 mm mesh’s end effect [12]); and the lead field between two interior points is reciprocal to 1.35%. Two finite-element items stay open (Sec. 6): on FEM  $\varphi$  the SFAP amplitude is erratic at  $\pm 40\%$  across depth while shape, timing and spectrum are faithful, and the FEM cylinder decays  $\approx 30\%$  slower with depth than the analytical one.

### 4.3 MUAP phenomenology

Each check is run on our pipeline and on the ported Farina generator, with arrays and montages built by translating the lead field (Figs. 8, 9). Between the junction and the tendon the monopolar SFAP is  $+ - +$  with the main lobe negative [36]. Conduction velocity from a single-differential array recovers 3.13, 3.98, 4.88 m/s for set values 3, 4, 5 [37]. Monopolar potentials mirror about the junction ( $r > 0.998$ ); the single-differential channel over it is 4% of the maximum and the channels either side are mirror images of opposite sign [38]. The end-of-fibre potential has the same onset on every electrode; its ratio to the propagating part



**Figure 8:** Physical time on a finite-element lead field (fibre 10 mm deep, tendons at  $\pm 60$  mm,  $v = 4$  m/s). (a) Monopolar SFAPs along a nine-electrode array with hard tendons: the propagating lobe walks at  $|z|/v$  (a fit gives 4.04 m/s; the guide is drawn at  $|z|/v + 1.0$  ms because the negative phase trails the wave front by the action potential’s rise) while the end-of-fibre potential starts at 15.4–15.6 ms on every channel ( $L/v = 15$  ms; spread 0.24 ms). (b) The single-differential montage vanishes and reverses at the innervation zone. (c) The same array with the one-sided tendon window of the direct recipe: the end-of-fibre potential falls from 0.16 to 0.10 of the propagating amplitude.

grows monotonically with depth and is suppressed mono  $>$  SD  $>$  DD [39, 40]. Amplitude is a power law in depth over 7–25 mm ( $R^2 > 0.98$ ) [27, 41]; RMS falls with subcutaneous fat as in the finite-element curve of Kuiken et al. [42]; amplitude and mean frequency fall with electrode radius [36]; with a single travelling wave the two electrodes of a bipolar pair see shift-copies to  $2 \times 10^{-3}$ , so the montage is the textbook comb filter with nulls at  $n v/\text{IED}$  [35, 43]; anisotropy elongates the lead field along the fibres by  $\sqrt{5}$  in the infinite medium and by 1.5 in the layered cylinder; and 50 identical fibres give  $50\times$  the SFAP to  $10^{-15}$ . The numbers behind these are the material of Sec. 5.1.

#### 4.4 Interference EMG and the pool

On the released forearm pool the rates lie in 5–40 pps with the onion-skin ordering ( $\rho = -0.98$  between threshold and rate); inter-spike-interval variability is 0.16; thresholds are right-skewed and twitches span 95 $\times$ ; the amplitude distribution is super-Gaussian at low drive (kurtosis 5.6 at 5%) and Gaussian at high (2.9–3.1 at 50–90%), with ARV/RMS 0.77–0.79, between the Laplacian and Gaussian values [44, 45]; the RMS at half maximal force is 0.56 of that at maximal force, inside the linear-to-quadratic band of Lawrence and De Luca [46]; larger units give larger MUAPs ( $\rho = 0.98$ ). The chain-level checks pass: orderly recruitment, onion skin, EMG rising with drive, force monotone and calibrated to 100% MVC at full drive, a localised MUAP footprint, an 8.7 $\times$  amplitude modulation across a movement, and a MUAP scale inside the physiological band (median peak-to-peak 24  $\mu\text{V}$ , range 0.7–160  $\mu\text{V}$ , median duration 18 ms, against tens to hundreds of microvolts and 5–20 ms [36]). Conduction velocity read off the HD grid is 4.33 m/s (4.1–4.9 over the 100 units) for a set 4.0 m/s: the 8–24% overestimate is the apparent velocity  $v/\cos\theta$  of fibres running 5–25 $^\circ$  to the grid columns (the same estimator recovers 3.98 on the cylinder; inclination bias of this size is documented [47]). The Gaussian renewal process has no refractory floor (1.6% of intervals below 20 ms at 40 pps) — the one tier-C failure.



## 5 Studies with the pipeline

The pipeline is a tool; this section uses it. Each study asks one question a user of surface EMG might ask, answers it with the pipeline, and ends with what to take from it.

### 5.1 What the electrode sees: depth, fat, spacing, montage

On the validation cylinder ( $v = 4$  m/s; fibre 10 mm deep unless stated) the peak-to-peak amplitude of a fibre falls with depth as a power law with exponent 2.9 monopolar and 3.5 single-differential over 7–25 mm (Fig. 9a): a fibre at 20 mm gives  $\approx 8\%$  of the monopolar amplitude of one at 7 mm and  $\approx 3\%$  of the single-differential one. Subcutaneous fat attenuates the RMS to 0.50, 0.23 and 0.12 at 3, 9 and 18 mm (Fig. 9b) and lowers the mean frequency. The end-of-fibre potential — the non-propagating component every electrode sees at the same instant — grows from 0.02 of the propagating amplitude at 7 mm to 0.20 at 20 mm (Fig. 9c), and the montage suppresses it: mono 0.053, single-differential 0.014, double-differential 0.004 at 10 mm. A larger electrode low-passes the signal ( $-2.8$  dB at 100 cycles/m for a 5 mm disc); the single-differential amplitude doubles from an inter-electrode distance of 2.5 to 5 mm and saturates beyond 10 mm, and the bipolar montage is a comb filter with nulls at  $n v/\text{IED}$ . Across the skin, the amplitude of a fibre 15 mm deep falls to half at 45 mm monopolar and 41 mm single-differential from the fibre (Fig. 9f).

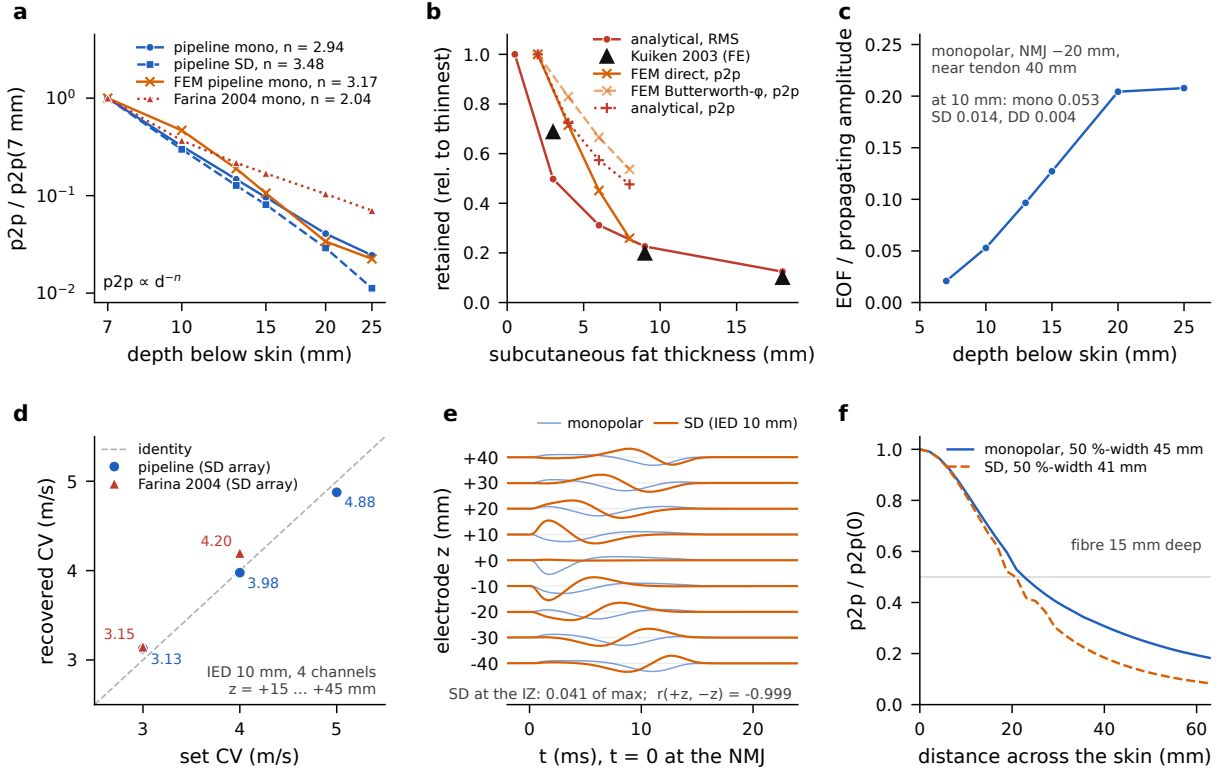
*What to take from it.* Surface EMG is a shallow measurement: the top 10 mm of a muscle dominate every montage, and units below  $\approx 20$  mm are represented mostly by their end-of-fibre potential, which carries no conduction-velocity information and is what differential montages remove. For selectivity, double-differential at 5–10 mm spacing removes most of the far-field component at the cost of amplitude; for amplitude, monopolar or single-differential at  $\geq 10$  mm spacing. Fat thickness should be reported with any amplitude comparison across subjects: 9 mm of fat costs 77% of the RMS.

### 5.2 Does fibre geometry change the signal?

We re-drew 24 units of the released FCU pool (5–395 fibres, 4–16 mm deep, including the three units of Fig. 4 and the largest unit) on the harmonic-streamline bed with the same territory centres and sizes, the same innervation plane and the same grid, and synthesised their MUAPs with the same recipe on the column of largest amplitude (Fig. 10). For the 18 units whose streamlines span the muscle, geometry is a second-order effect: peak-to-peak amplitude changes by a median factor of 1.01 (0.86–1.18) with no dependence on depth or size, the waveforms correlate at  $r = 0.98$  (0.88–1.00) with lags below 0.5 ms, the harmonic MUAPs are  $\approx 2$  ms shorter with a  $\approx 20\%$  smaller end-of-fibre component (curved paths reach the tendons at spread-out times), and the conduction velocity read off the column agrees to 3% (4.28 against 4.30 m/s). The first-order effect is not geometry but innervation. A sixth of the FCU streamlines are truncated — they start distal to the innervation plane, on the deep side of the muscle — and the single-junction rule places their junction at the fibre’s end; the six units that draw on them acquire a synchronous, non-propagating onset potential  $0.7\text{--}60\times$  the propagating lobe, a  $2\text{--}19\times$  inflated amplitude,  $r \approx 0$  and no usable conduction velocity (unit L in Fig. 10c). Re-drawing those units from full-span streamlines removes the artefact ( $r \geq 0.96$ ). At the fibre level the beds differ where expected: streamline lengths are bimodal (median 193 mm, a fifth below 120 mm) against one global 204 mm, and containment is 99.9% against 96.6%; but — a correction to the usual description — the straight bed is not constant-depth either, since its paths follow the muscle centreline (depth excursion 6.4 mm along a straight fibre, 5.2 mm along a streamline).

*What to take from it.* For HD-EMG synthesis the straight bed is an adequate default: the same amplitudes, shapes and velocities within a few percent. Use the harmonic bed when anatomy matters — fibres that stay inside the muscle and follow its shape — and prune or re-





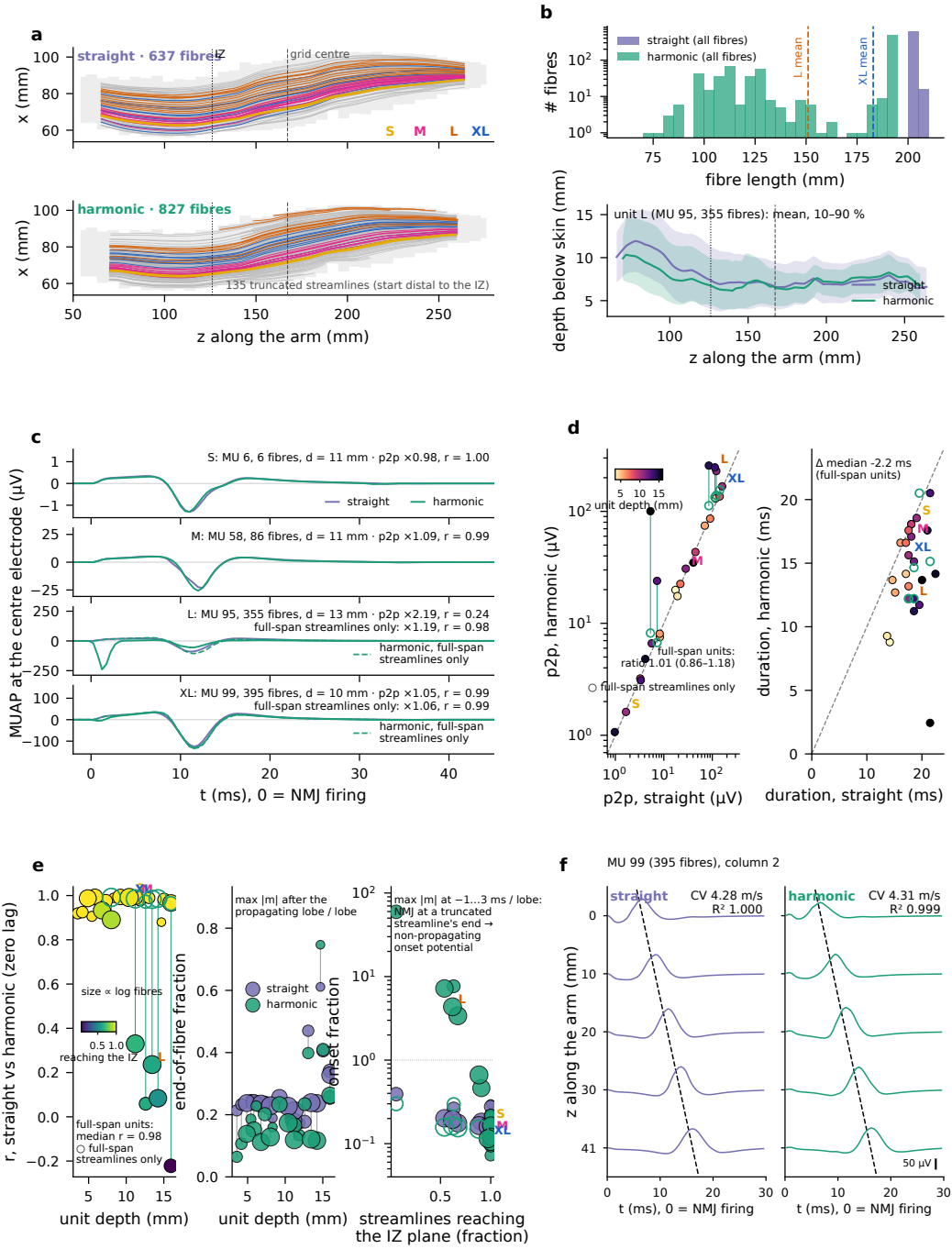
**Figure 9:** MUAP features against the literature (fibre 10 mm below the skin unless stated;  $v = 4$  m/s). (a) Depth law of the peak-to-peak amplitude (log-log) for the pipeline (monopolar and single-differential), the Farina generator and the FEM pipeline, with fitted exponents. (b) RMS retained against subcutaneous fat thickness, analytical model against the finite-element points of Kuiken et al. [42]; FEM pipeline (direct and Butterworth-smoothed) and Farina generator at 2–8 mm, each relative to its thinnest case. (c) End-of-fibre to propagating amplitude ratio against depth (junction 20 mm and near tendon 40 mm from the electrode). (d) Conduction velocity recovered from a four-channel single-differential array against the set value. (e) The monopolar and single-differential array across the innervation zone (IED 10 mm). (f) Transverse spread of monopolar and single-differential amplitude for a fibre 15 mm deep.

innervate truncated streamlines first, because a junction at a fibre’s end is a far-field generator that no montage removes. Neither bed narrows the surface field to real motor-unit widths; that is the job of the series-fibered variant, not of curvature.

### 5.3 Crosstalk: what a grid over one muscle receives from its neighbours

The same 25 lead fields serve every muscle in the segmentation, so this question costs no new finite-element work. We built straight beds and size-principle pools for the four muscles around the FCU — flexor digitorum superficialis (FDS; lateral and superficial, its edge 4 mm from the grid’s outer column), flexor digitorum profundus (FDP; deep, directly beneath the FCU), extensor carpi ulnaris (ECU) and flexor carpi radialis (FCR; on the far sides of the ulna and of the FDS) — and synthesised seven units per muscle (5–395 fibres) on the FCU grid (Fig. 11). Crosstalk is expressed as the ratio  $R$  of a neighbouring unit’s largest amplitude on the grid to what an FCU unit of the same size gives (the FCU size law of the 100-unit pool,  $r = 0.97$ ).

Monopolar,  $R$  does not fall with distance: it is 0.05–0.25 for every unit from 17 to 55 mm from the grid centre (fitted slope indistinguishable from zero,  $r = -0.16$ ); per muscle the medians are 0.07 (FDS), 0.17 (FDP), 0.19 (ECU) and 0.16 (FCR), and 22 of the 28 units exceed 10%. What reaches the grid from the far muscles is a single positive lobe at  $\approx 8$  ms — the non-propagating end-of-fibre far field — which is why it neither walks along the grid nor fades with distance. The largest ECU and FCR units, 46–51 mm around the arm, put 35 and 30  $\mu$ V on the grid and 19



**Figure 10:** Straight against harmonic-streamline fibre beds in the FCU: same units, innervation plane, grid and recipe. (a) Side view of the two beds with the S/M/L/XL units highlighted; 135 of the 827 streamlines start distal to the innervation plane. (b) Fibre lengths (streamlines bimodal; straight fibres one length) and depth below the skin along the fibres of unit L (mean and 10–90% band): both beds vary in depth along the fibre. (c) MUAPs of the four units at the centre electrode, straight and harmonic; for L and XL also the harmonic unit re-drawn from full-span streamlines. (d) Peak-to-peak amplitude and duration, harmonic against straight, colour by depth; open circles are the full-span re-draw. (e) Waveform correlation, end-of-fibre fraction and onset fraction (largest excursion in the first 3 ms relative to the propagating lobe) against unit depth or the fraction of the unit's streamlines that reach the innervation plane: the units off the diagonal are the truncated-streamline cases. (f) Column waterfall of the largest unit on both beds with the conduction velocity read off the grid.

and  $13\ \mu\text{V}$  at its centre — more than the largest FDP unit directly beneath the FCU ( $16\ \mu\text{V}$  on the grid,  $3.6$  at the centre), because depth attenuates a field faster than distance around a limb does. The FDS is a different case: its largest unit’s territory lies under the grid’s outer column and its innervation zone under the fourth row, so it reaches  $0.76$  of the FCU value monopolar and  $0.63$  in either differential montage — not crosstalk a montage can remove, but a grid that covers two muscles. Differential montages change the picture. Relative to monopolar, single-differential keeps a median  $0.34$  of the neighbours’ amplitude ratio ( $0.19$  of their energy) and double-differential  $0.24$  ( $0.06$ ), while for the FCU’s own units the same montages keep  $1.0$  and  $1.6$  of the energy, so the own-to-crosstalk energy ratio improves  $5.5\times$  and  $28\times$ ;  $R$  now falls with distance ( $r \approx -0.6$ ), crosses  $10\%$  at  $24\ \text{mm}$  (single-differential) and  $19\ \text{mm}$  (double-differential), and no unit beyond  $30\ \text{mm}$  exceeds it (monopolar:  $55\ \text{mm}$ ).

*What to take from it.* A monopolar grid over a forearm muscle is not selective for that muscle: units of a superficial neighbour  $50\ \text{mm}$  away appear at  $10\text{--}20\%$  of the amplitude of an own unit, and distance alone does not remove them. Differentiation does, at no cost to the own units — the design rule is a differential montage, and a grid narrow enough not to overlap the neighbour’s territory. The pipeline gives these numbers for any muscle and any layout in the segmentation for the price of one lead-field solve per electrode. Two caveats: the monopolar potentials are referenced to the volume mean of the conductor rather than to a remote electrode, so the monopolar far-field ratios are reference-dependent (the differential ones are not); and all five muscles are cut by the imaged volume, so the straight beds span the field of view rather than anatomical fascicle lengths.

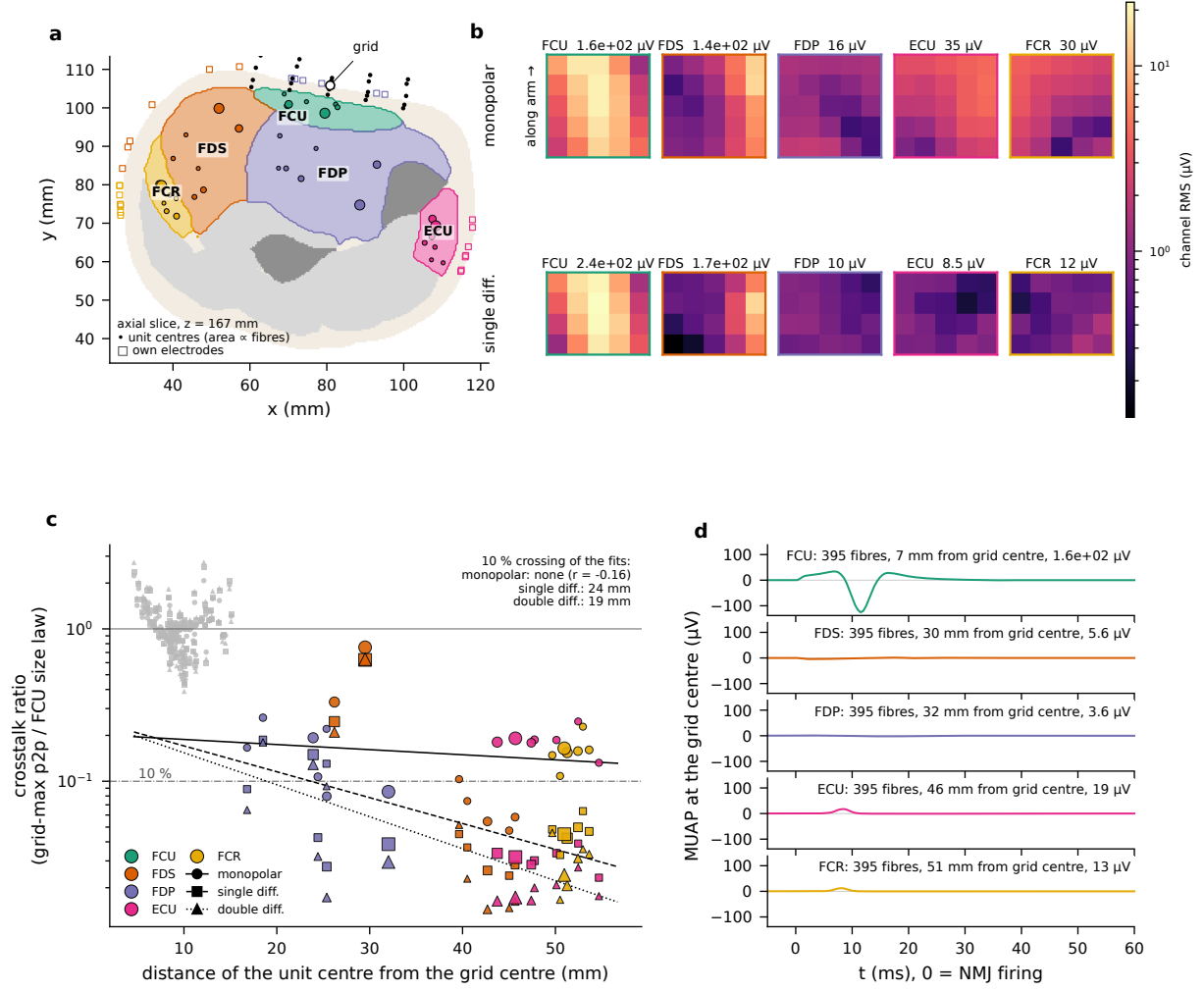
## 5.4 Interference EMG across drive

Six trapezoid contractions of the forearm pool at drives  $0.1\text{--}1.0$  (dataset D3) give plateau RMS of  $14, 27, 47, 73, 101$  and  $133\ \mu\text{V}$  at the centre electrode for forces of  $7, 16, 30, 46, 69$  and  $100\%$  MVC, with  $53, 68, 81, 90, 98$  and  $100$  of the  $100$  units active. The EMG–force relation is between linear and quadratic (RMS at half force  $0.56$  of that at full force), where Lawrence and De Luca [46] place the forearm muscles. The amplitude distribution moves from super-Gaussian at low drive to Gaussian at high, as the number of superimposed trains grows [44], and the median frequency at moderate drive is  $77\ \text{Hz}$ . Amplitude cancellation — the loss of RMS when MUAPs of opposite sign overlap — is  $68\%$  already at  $20\%$  drive, twice the  $\approx 33\%$  that Keenan et al. [48] report at that level; a movement trial modulates the RMS  $8.7\times$  between flexed and extended.

*What to take from it.* The pool reproduces the shape of the EMG–force curve and the statistics of the interference signal without tuning, so the drive can be read from the RMS with the literature’s calibration. Cancellation is the one statistic the pool overshoots, and it tells us something about the fibre model:  $100$  units firing  $18\ \text{ms}$  MUAPs from shared, full-length fibres overlap more than a real pool’s do, because their surface fields are broader than real units’ — a property of fibre length and the single junction, not of curvature (Sec. 5.2). Absolute RMS values carry the  $\sigma_{\text{in}}\pi a^2$  caveat of Sec. 3.5 and should be compared within the model, not across to recordings.

## 6 Discussion and outlook

`emgforge` turns a segmentation into surface EMG with no hand steps, and every stage in between is a checkable object. The two decisions that shape it are reciprocity, which makes the cost proportional to electrodes rather than fibres, muscles or contractions, and direct line-source synthesis, which evaluates the physics of the SFAP directly on the lead field and is checked against a closed-form solution. The validation suite found the model’s own mistakes along the way — a spurious  $1/v$  in the amplitude, an electrode placement that had been misread as a



**Figure 11:** Crosstalk on the FCU grid from the neighbouring muscles. (a) The forearm slice at the grid centre: the FCU and its four neighbours, the seven unit centres per muscle (area  $\propto$  fibres), the grid, and one “own” skin electrode per unit (squares). (b) Monopolar (top) and single-differential (bottom) RMS maps on the grid for the largest unit (395 fibres) of each muscle, shared log colour scale, peak-to-peak amplitude above each map. (c) Ratio  $R$  of each neighbouring unit’s largest amplitude on the grid to the FCU size law at its size, against distance from the grid centre, for the three montages (log-linear fits and their 10% crossings); grey: the FCU units themselves. (d) MUAPs at the grid centre for the largest unit of each muscle at a shared scale, with each unit’s distance from the grid centre.

model property, and the anti-phase behaviour of our spatial-frequency port — and we list what remains open rather than smooth it over:

- *Conduction velocity and amplitude.* The action potential is fixed in space, so amplitude is independent of  $v$ ; physiologically it is fixed in time and a faster fibre carries a longer wave, so single-fibre amplitude should fall with  $v$  [6].
- *Finite-element amplitude and depth law.* On FEM  $\varphi$  the SFAP amplitude is erratic at  $\pm 40\%$  across depth because the recipe integrates  $\varphi''$  faithfully down to  $\approx 8$  mm wavelengths, where FEM  $\varphi$  carries mesh structure the three-pole fit does not remove; low-passing  $\varphi$  smooths the amplitudes but halves the mean frequency (46 vs 112 Hz). The FEM cylinder also decays  $\approx 30\%$  slower with depth than the analytical one. A finer mesh around the fibre band, and a refinement study of the  $\approx 47,000$ -cell forearm mesh, are the next numerical work.
- *Fibre model.* Full-length, single-junction fibres — straight or curved (Sec. 5.2) — give

**Table 5:** Datasets released with the paper (float32, NumPy `.npz` with a JSON manifest each; 8 MB in total).

| name                                 | contents                                                                                                                                                                                                                | size   |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <code>cylinder_sfap_atlas</code>     | analytical and FEM lead fields and SFAPs on the validation cylinder for 6 depths $\times$ 5 angles $\times$ 3 junction offsets $\times$ 9 electrode positions (1620 SFAPs), with the Farina-generator MUAP of each case | 1.6 MB |
| <code>forearm_fcu_mu_pool</code>     | the 100-unit FCU pool (fibre paths, fibre indices, sizes, territories, activation parameters), MUAPs at the centre electrode and the full (100, 25, 256) HD-grid MUAP tensor, electrode positions, configuration        | 2.4 MB |
| <code>interference_emg_trials</code> | trapezoid contractions at six drive levels (0.1–1.0): drive, spike trains, single-channel and 25-channel EMG, force; one movement trial                                                                                 | 4.1 MB |

surface fields broader than real units’; the largest units stay below the 1–2 mV of the largest superficial recorded units and cancellation is overshot. The series-fiberling variant of the harmonic bed addresses this and awaits validation.

- *Spatial-frequency port.* Anti-phase and lagged against the integral; to be audited for the action-potential orientation and sign conventions.
- *Pool.* The Gaussian renewal process has no refractory floor.
- *No experimental confrontation yet.* The next validation is spike-triggered surface MUAPs from decomposed HD-EMG on the same subject against the MRI pipeline, with the error and correlation targets of Lowery et al. [13], Botelho et al. [14].

Three uses are immediate. The interference-EMG trials carry ground-truth spike trains, unit properties and a full provenance record, which is what a decomposition benchmark such as MUniverse [17] consumes; an exporter to its BIDS-based layout is the natural next step. The lead fields of the parametric limb are the training data for neural-operator surrogates [20], and the validation tiers give those surrogates a signal-level target (the SFAP, not only the field). And a forward model that is exact against first principles is the reference that model-informed decomposition [18] needs when it inverts a learned MUAP generator. Because the pipeline takes any segmentation, the same studies can be repeated on other muscles, limbs and subjects without new code; that, more than any single number here, is what it is for.

## 7 Released datasets

Three datasets are generated by scripts in `paper/figures/` and described by manifests in `paper/datasets/` (Table 5). They are reference material: every array can be regenerated from the repository.

## Code and data availability

Code: <https://github.com/dhalatsis/emgforge> (MIT), branch `paper/arxiv-v1`. The end-to-end runner is `scripts/run_pipeline.py`; the validation suite is `scripts/validation/run_all.py`; the figure, study and dataset scripts are in `paper/figures/`; dataset manifests in `paper/datasets/`.

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## A The validation checks

Table 6 lists every check of the suite with its criterion and the measured result, generated from the suite’s JSON records (`paper/make_checks_table.py`).

**Table 6:** Every check of the validation suite: what it asserts, its criterion (first clause), what was measured (first clause), and the outcome. Full strings are in `docs/validation/REPORT.md`.

| check                                                                 | criterion | measured | result |
|-----------------------------------------------------------------------|-----------|----------|--------|
| <i>Tier A — cylinder: first principles, analytical, FEM, pipeline</i> |           |          |        |

| check                                                                                                              | criterion                                                                                                                                                | measured                                                                                                                                                                      | result |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A0.1 spatial engine $\equiv$ first-principles line-source integral (shape, timing)                                 | $r \geq 0.999$ and $ \text{lag}  \leq 0.1$ ms: same integral, so only discretisation can differ                                                          | $r = [1.0, 0.9999, 0.9999]$ , $\text{lag}(\text{ms}) = [-0.0, -0.05, -0.05]$ over NMJ offsets 0/-20/-30 mm                                                                    | pass   |
| A0.2 spatial engine amplitude constant equals first principles                                                     | ratio $1.00 \pm 5\%$ , independent of $v$                                                                                                                | engine/reference amplitude ratio = 0.994 at $v=2$ , 0.994 at $v=4$                                                                                                            | pass   |
| A0.3 Fourier engine vs first principles (signed $r$ , lag)                                                         | $r \geq +0.99$ (same integral, same sign, no lag)                                                                                                        | signed $r = [-0.795, -0.684, -0.643]$ at lag $[1.4, 2.55, 0.85]$ ms; vs a spatially MIRRORED IAP: $r = [-0.788, -0.684, -0.642]$                                              | known  |
| A0.4a sampling invariance: fsamp 4096 $\rightarrow$ 2048 and dz 0.98 $\rightarrow$ 0.5 mm leave the SFAP unchanged | $r > 0.995$ , amplitude within 2 % (discretisation-converged)                                                                                            | fsamp: $r=1.00000$ p2p ratio=0.9977; dz: $r=0.99886$ p2p ratio=1.0011                                                                                                         | pass   |
| A0.4b translation: moving the electrode $\Delta z$ along the fibre delays the propagating lobe by $\Delta z/v$     | $ \Delta  \leq$ one sample (0.24 ms)                                                                                                                     | $\Delta z=20$ mm $\rightarrow$ main-lobe delay 4.88 ms (expect 5.00)                                                                                                          | pass   |
| A0.5 the CSD source is monopole-free: $\int i_m(z,t) dz = 0$ at every instant                                      | $< 1e-6$ : generation and end-of-fibre terms exactly balance the propagating tripoles                                                                    | $\max_t  \Sigma_z i_m  / \Sigma_z  i_m  = 7.7e-17$ (boxcar), $7.7e-17$ (one_sided)                                                                                            | pass   |
| A0.4c superposition: a two-fibre MUAP is the sum of its SFAPs                                                      | $< 1e-9$ (linear volume conductor)                                                                                                                       | $\max \text{MUAP} (\text{SFAP}_1+\text{SFAP}_2)  / \text{p2p} = 0.0e+00$                                                                                                      | pass   |
| A0.4d CV scaling: $v$ 4 $\rightarrow$ 3 stretches the SFAP by 4/3 and scales its spectrum by 3/4                   | both within 8 %                                                                                                                                          | duration ratio 1.328 (expect 1.333), MNF ratio 0.757 (expect 0.750)                                                                                                           | pass   |
| A1.1 FEM $\varphi(z)$ reproduces the analytical cylinder $\varphi(z)$ along the fibre (5 depths)                   | $r \geq 0.99$ ; FWHM within 20 % (electrode models differ: $\varnothing 10$ mm disk vs $\sigma=5$ mm Gaussian blob, so the shallowest fibres see...      | $r = [0.9962, 0.9948, 0.9969, 0.9984, 0.9998]$ ; FWHM $_z$ ana/FEM (mm) = ['41.0/35.1', '53.1/45.3', '63.1/57.7', '68.9/65.8', '81.2/82.1'] at $r = 33/30/27/25/20$ mm        | pass   |
| A1.1b ... and its second derivative $\varphi''(z)$ — the kernel the SFAP actually integrates                       | $r \geq 0.98$ at the best-matched electrode radius (the FEM electrode is a $\sigma=5$ mm Gaussian blob clipped by the skin — the analytical disk...      | $r(\varphi'')$ vs analytical electrode radius 2.5/5/7.5/10 mm: $r=33$ : [0.985, 0.964, 0.916, 0.842]; $r=27$ : [0.931, 0.923, 0.909, 0.889]; $r=20$ : [1.0, 1.0, 1.0, ...]    | fail   |
| A1.2 FEM/analytical SFAP amplitude ratio is one constant across depth (same depth law)                             | TWO findings. (i) A smooth 1.3–1.4 $\times$ drift survives every preprocessing and a $\sigma=1$ mm source: the FEM cylinder decays $\sim 30\%$ slower... | p2p ratio FEM/ana, $r=33\rightarrow 20$ mm, rel. to $r=33$ — direct method: [1.0, 1.46, 1.35, 1.2, 0.86] (spread 1.69 $\times$ ); no denoise: [1.0, 1.59, 1.3, 1.72, 2.03]... | pass   |
| A1.3 FEM $\varphi(z)$ vs analytical for fibres off the electrode meridian ( $\theta = 10\text{--}45^\circ$ )       | $r \geq 0.99$                                                                                                                                            | $r = [0.9978, 0.9997, 0.9985, 0.997]$                                                                                                                                         | pass   |
| A2.1 pipeline on FEM $\varphi \equiv$ pipeline on analytical $\varphi$ (4 depths $\times$ 2 NMJ offsets)           | $r \geq 0.95$ with the direct recipe; deep fibres $\geq 0.99$ . The shallow-fibre residual is the FEM field (electrode model + mesh ripple through...    | direct method: $r = [0.948, 0.97, 0.9, 0.929, 0.946, 0.915, 0.998, 0.995]$ (depth 33,33,30,30,25,25,20,20 mm; no denoise: [0.948, 0.954, 0.891, 0.909, 0.919, ...]            | known  |

| check                                                                                                   | criterion                                                                                                                                       | measured                                                                                                                                                                 | result |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A2.2 monopole denoise removes FEM mesh ripple without changing the analytical answer                    | denoised jaggedness $\leq$ raw; agreement not reduced                                                                                           | SFAP jaggedness<br>raw→denoised =<br>[‘0.004→0.003’, ‘0.007→0.002’,<br>‘0.006→0.003’, ‘0.011→0.002’]<br>...                                                              | pass   |
| A2.3 end-of-fibre onset lands at L/v in the FEM pipeline (and where the Farina generator puts it)       | pipeline $ \Delta  \leq 0.5$ ms                                                                                                                 | expected/Farina/pipeline (ms): 10.0/7.57/10.26, 15.0/12.45/15.39 → pipeline $ \Delta  \leq 0.39$ ms; Farina onset leads by up to 2.55 ms (its IAP body...                | pass   |
| A2.4 conduction velocity recovered from a 4-channel longitudinal array (both models)                    | within 5 % of the set CV, linear lag-distance $R^2 > 0.99$                                                                                      | Farina 3.83 m/s ( $R^2=0.998$ ); FEM pipeline 3.98 m/s ( $R^2=0.999$ ); set 4.0                                                                                          | pass   |
| A2.5 waveform $ r $ vs the Farina generator (sign-agnostic, $\pm 6$ ms lag)                             | $ r  \geq 0.9$ — currently limited by the IAP-orientation/sign conventions of the Fourier/Farina family (A0.3)                                  | spatial/Fourier: -0.91/+0.94, -0.74/+0.74, -0.93/+0.52 at NMJ 0/-20/-30 mm                                                                                               | known  |
| A3.1 rotational symmetry: electrode $+30^\circ \equiv$ fibre $-30^\circ$                                | $r > 0.999$ , peak within 3 % (mesh-discretisation level)                                                                                       | $r = [1.0, 1.0]$ , peak ratio = [0.9986, 1.0109]                                                                                                                         | pass   |
| A3.2 axial translation: electrode $+20$ mm $\equiv$ fibre window $-20$ mm (Neumann end effect)          | $r > 0.995$ , peak within 3 % — the residual ( $\leq 0.5$ %) is the finite-length (240 mm) mesh’s end effect                                    | $r = [0.99865, 0.99561]$ , peak ratio = [0.9984, 0.9946] (electrode 100 mm from the mesh end)                                                                            | pass   |
| A3.3 reciprocity between two interior points in the anisotropic muscle                                  | ratio $1 \pm 5$ % (symmetric $\sigma \Rightarrow$ Green’s function symmetric)                                                                   | $\varphi\_A(B) = 2.3625e-02$ , $\varphi\_B(A) = 2.3311e-02$ , ratio 1.0135                                                                                               | pass   |
| <i>Tier B — MUAP features against the literature</i>                                                    |                                                                                                                                                 |                                                                                                                                                                          |        |
| B1 monopolar SFAP over the fibre (between IZ and tendon) is triphasic + - + with the main lobe negative | + - + with the largest lobe negative (depolarised zone under the electrode). The Farina port’s propagating main lobe is POSITIVE: its output... | pipeline lobes [1, -1, 1] (largest -1); Farina generator lobes [-1, 1, -1, 1, -1] (largest +1); electrode 30 mm from the NMJ, tendons $\geq 95$ mm away,...              | pass   |
| B2 conduction velocity recovered from an SD array tracks the set CV (3, 4, 5 m/s)                       | within 5 % (pipeline) / 6 % (Farina) at each CV; physiological range 3-5 m/s                                                                    | pipeline/Farina: v=3: 3.13/3.15 m/s, v=4: 3.98/4.20 m/s, v=5: 4.88/nan m/s (Farina NaN = its k-grid does not build at that v)                                            | pass   |
| B3 innervation-zone signature: monopolar mirror symmetry, SD null and phase reversal at the IZ          | mono $r > 0.99$ ; SD at the IZ $< 5$ % of max; SD $r(+d, -d) < -0.95$ (mirror images of opposite sign)                                          | pipeline: mono $r(+d, -d) = [0.9981, 0.9979, 0.9985]$ , SD@IZ/max = 0.041, SD $r(+d, -d) = [-0.999, -0.998, -0.998]$ ; Farina: SD@IZ/max = 0.004, SD $r(+d, -d) = \dots$ | pass   |
| B4a the end-of-fibre component is non-propagating: same onset on every electrode                        | spread $< 0.5$ ms across electrodes $-30 \dots +30$ mm                                                                                          | pipeline onsets [10.51, 10.51, 10.51, 10.51, 10.51] ms (spread 0.00); Farina [7.57, 7.81, 7.57, 7.57, 7.57] (spread 0.24); expected L/v = 10 ms                          | pass   |
| B4b EOF / propagating amplitude ratio increases monotonically with depth                                | monotone increase (the propagating part decays faster than the far-field EOF)                                                                   | ratio at depth-below-skin 7/10/13/15/20 mm = [0.021, 0.053, 0.097, 0.127, 0.204]                                                                                         | pass   |

| check                                                                                                          | criteria                                                                                                                                                   | measured                                                                                                                                                                                        | result |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| B4c spatial filters suppress the EOF/propagating mono > SD > DD                                                | strict ordering                                                                                                                                            | {'mono': 0.053, 'SD': 0.014, 'DD': 0.004} (IED 10 mm)                                                                                                                                           | pass   |
| B5 amplitude vs depth is a power law (log-log linear), steeper for SD than monopolar                           | $R^2 > 0.95$ over 7–25 mm; $n_{SD} > n_{mono}$ ; monotone decrease. Note the Farina generator's own exponent differs from the first-principles engine's... | exponent $n$ ( $p2p \propto d^{-n}$ ), $R^2$ : pipeline mono 2.94 (0.998), SD 3.48 (0.998); Farina generator mono 2.04 (0.986); FEM pipeline mono 3.17 (0.987)...                               | pass   |
| B6 transverse spread: monopolar wider than SD (lit. 2.5–4 $\times$ ); depth $\approx 0.2 \times$ (mono 50...   | $mono \geq SD$ width; both widen with depth. The literature ratio (2.5–4 $\times$ ; biceps MUs 15–25 mm deep, mono 72–96 mm vs SD 24–32 mm) comes...       | 50 %-width (mm) at depth 15/20 mm: mono 45/88, SD 41/53 (ratio 1.1/1.7); depth/(0.2-width) = 1.66/1.13 (Roeleveld's biceps-MU rule = 1.0; single fibre...                                       | pass   |
| B7 MUAP amplitude $\propto$ fibre count at fixed geometry; NMJ scatter ( $\sigma=5$ mm) disperses and...       | exact linearity; sub-linear amplitude and longer duration with dispersion                                                                                  | 50 identical fibres = 50 $\times$ SFAP to 9.9e-16; with NMJ $\sigma=5$ mm: p2p 0.86 $\times$ the coherent sum, 10 %-duration 20.0 $\rightarrow$ 22.2 ms                                         | pass   |
| B8 the bipolar montage is a pure spatial difference of a translation-invariant wave ( $\Rightarrow$ comb...    | equivariance $< 5e-3$ : $SD(t) = S(t) - S(t - IED/v)$ to that precision, so $ H  =  2 \sin(\pi f \cdot IED/v) $ with nulls at $n \cdot v / IED$ follows... | translation equivariance residual (single travelling wave, 9–33 ms): no denoise 1.4e-03, direct method 2.2e-03. Measured null of $ SD / mono $ — pipeline (no...                                | pass   |
| B9 electrode size is a spatial low-pass: amplitude and MNF fall with electrode area; $\varnothing 5$ mm...     | monotone decrease; -3 dB points as in the literature                                                                                                       | Farina p2p vs radius (0.5, 2.5, 5.0, 10.0): [1.0, 0.98, 0.924, 0.765]; MNF: [81.7, 81.0, 78.8, 72.8] Hz; disc filter: $\varnothing 5$ mm @100 c/m = -2.8 dB,...                                 | pass   |
| B10 SD amplitude grows $\approx$ linearly with IED when $IED \ll \lambda$ , then saturates towards $\lambda/2$ | differentiator regime then saturation ( $\lambda \approx v \cdot T_{lobe} \approx 20$ –30 mm)                                                              | SD p2p vs IED (2.5, 5.0, 10.0, 20.0) mm — pipeline: [1.0, 1.95, 3.59, 5.28]; Farina: [1.0, 1.95, 3.52, 5.01] ( $\times 2$ IED $\rightarrow \geq 1.7 \times$ at small...                         | pass   |
| B11 subcutaneous fat attenuates (Kuiken 2003: -31/-80/-90 % at 3/9/18 mm) and low-passes the surface...        | within $\times 1.5$ of Kuiken's FE curve; RMS monotone decreasing and MNF lower at 3 and 9 mm than at 0.5 mm; FEM (smoothed $\varphi$ ) vs analytical...   | analytical RMS retained at fat (0.5, 3.0, 6.0, 9.0, 18.0) mm: [1.0, 0.498, 0.312, 0.226, 0.124] (Kuiken 0.69/0.20/0.10 at 3/9/18); MNF [91.0, 79.0, 76.0, 79.0, ...                             | pass   |
| B12 muscle anisotropy ( $\sigma_z/\sigma_r = 5$ ) elongates the lead field along the fibres                    | $\sqrt{5}$ exactly in the infinite medium; 1.3–3 in the layered cylinder (fat/skin layers and the bounded cross-section compress the elongation); FEM...   | FWHM <sub>z</sub> (aniso)/FWHM <sub>z</sub> (iso)pass analytical cylinder 1.48, infinite medium 2.236 (theory $\sqrt{5} = 2.236$ ); FEM along/across FWHM at $r=30$ : 2.29 (layered, bounded... | pass   |

*Tier C — interference EMG and motor-unit pool*

|                                                                                              |                                                                                                     |                                                                                                                                                              |      |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| C1 discharge rates lie in 5–40 pps and follow the onion skin (earlier-recruited fire faster) | model rates within 5–40 pps; measured within 10 % of nominal; $\rho(\text{threshold, rate}) < -0.9$ | model: min rate 5, peak 40 pps; measured @drive 0.5: 89 active, 7.6–36.5 pps, Spearman(threshold, rate) = -0.98; @drive 1.0 max 42.9 pps (renewal-process... | pass |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|



| check                                                                                                                       | criterion                                                                                                                                                 | measured                                                                                                                              | result      |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------|
| C2 inter-spike-interval variability: CoV 0.1–0.3, no ISI below the $\sim 20$ ms refractory floor                            | CoV $\in [0.1, 0.3]$ ; $< 1$ % of ISIs under 20 ms. The Gaussian renewal model has no refractory floor — at 40 pps (ISI 25 ms, $\sigma$ 4 ms) sub-20...   | median per-MU ISI CoV 0.16 (model ISI_CV = 0.167); ISIs $< 20$ ms: 1.58 %; min ISI 9.8 ms                                             | <b>fail</b> |
| C3 recruitment thresholds are right-skewed (many low, few high); twitch forces span $\approx 100\times$                     | skewness $> 0.5$ ; 50–200 $\times$ (Fuglevand RP = 100; FDI 130)                                                                                          | threshold skewness +1.25 (median/max = 0.12); twitch P_max/P_min = 95                                                                 | pass        |
| C4 amplitude PDF: super-Gaussian at low force, $\rightarrow$ Gaussian (kurtosis 3–3.5) at high; ARV/RMS $\in [0.71, \dots]$ | kurtosis decreasing with force to 2.8–3.6; ARV/RMS between Laplacian and Gaussian. NB: with the C-04 MUAPs (33 ms long) the low-force EMG is denser...    | kurtosis @drive 0.05/0.1/0.5/0.9 = [5.59, 4.56, 3.1, 2.92]; ARV/RMS @0.3/0.7 = [0.768, 0.786] (Laplacian 0.707, Gaussian 0.798)       | pass        |
| C5 amplitude cancellation grows with excitation ( $\approx 30$ % at 20 % $\rightarrow \approx 60$ % at maximum)             | increasing; 15–60 % at 0.2, 40–85 % at 1.0 (Keenan 2005: 33 % $\rightarrow$ 62–65 %). Cancellation is set by MUAP duration $\times$ active-MU count:...   | 1 – mean  $\Sigma$ trains  / $\Sigma$ mean train_i  @drive 0.2/0.5/1.0 = [68, 79, 81] %                                               | known       |
| C6 EMG–force relation lies between linear and quadratic: RMS at 50 % force = 0.3–0.6 of RMS at 100 %                        | 0.3 (quadratic, biceps/triceps) – 0.6 (linear, FDI/soleus); force monotone in drive. A ratio above 0.6 (EMG concave in force) follows from the...         | RMS(50 % MVC)/RMS(100 % MVC) = 0.56; force plateau %MVC = [7, 16, 30, 46, 69, 85, 100] at drive [0.1, 0.2, 0.35, 0.5, 0.7, 0.85, 1.0] | pass        |
| C7 interference-EMG spectrum: MNF/MDF at moderate force in 70–130 Hz                                                        | MDF 70–130 Hz (bipolar norms: biceps $90 \pm 18$ , TA $116 \pm 20$ ). Expected to fail while the C-04 fibre geometry yields 33 ms MUAPs — the spectrum... | @drive 0.5 (monopolar): MNF 81 Hz, MDF 77 Hz                                                                                          | pass        |
| C8 larger motor units produce larger MUAPs (size $\leftrightarrow$ amplitude on the recording side)                         | $\rho > 0.5$ (amplitude $\propto$ fibre count at fixed depth; depth scatter lowers $\rho$ )                                                               | Spearman(fibres per MU, grid-max p2p) = +0.98 over 100 MUs (sizes 5–395 fibres)                                                       | pass        |

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*Tier S — chain-level sanity*

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|                                                          |                                                                                                                        |                                                                                                                |      |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------|
| Orderly recruitment (Henneman size principle)            | n_active non-decreasing in drive; smallest units recruited first                                                       | n_active @ drive [0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0] = [37, 51, 67, 84, 94, 100, 100]; thresholds sorted=True | pass |
| Onion-skin rate coding                                   | earlier (smaller) units discharge faster than later (larger) ones                                                      | @drive 0.7: first-recruited 40.0 Hz vs last 5.6 Hz (98 active)                                                 | pass |
| Interference-EMG amplitude $\uparrow$ with contraction   | monotone increase with drive                                                                                           | plateau RMS( $\mu$ V) @ [0.1, 0.2, 0.35, 0.5, 0.7, 0.9] = [14.822, 28.279, 49.059, 61.24, 97.757, 124.795]     | pass |
| EMG $\leftrightarrow$ force: monotone and MVC-calibrated | monotone in drive; $\approx 100$ %MVC at full drive                                                                    | plateau force(%MVC) @ [0.1, 0.2, 0.35, 0.5, 0.7, 0.9, 1.0] = [7.4, 16.0, 30.3, 46.8, 69.8, 92.2, 102.5]        | pass |
| Conduction velocity from HD-EMG propagation              | physiological muscle-fibre CV, mean $\approx 4$ m/s — the model uses one fibre CV ( $\approx$ uniform across the pool) | pool median CV 4.32 m/s (range 4.1–4.8 over 100 MUs, IED 10.1 mm); corr(size, CV) = -0.41                      | pass |

| check                                      |   | criterion                                                                                                                                                      | measured                                                           | result |
|--------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------|
| Spatial selectivity of a MUAP on the array | a | a MUAP is localised on the array, not a uniform far field (ratio > 1.5)                                                                                        | peak-channel RMS / grid-mean RMS = 2.16 (MU 99)                    | pass   |
| Non-stationarity over a movement           | a | EMG amplitude is modulated by joint angle, not just by the drive                                                                                               | RMS flexed / RMS extended = 8.66 (90.856 vs 10.495 $\mu\text{V}$ ) | pass   |
| MUAP physiological scale                   |   | single-MU surface MUAP $\approx$ 20–800 $\mu\text{V}$ , 5–20 ms (Merletti & Muceli 2019; Farina 2014) (the 0.6 $\mu\text{V}$ / 33 ms of earlier runs was an... | median p2p 23.60 $\mu\text{V}$ , median duration 18.1 ms           | pass   |